A Reproducible Multi-Phase Literature-Review Pipeline for Exoplanet Atmospheres

Synthetic Fixture Exemplar with Source-Aware Retrieval and Optional LLM Analysis

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September 28, 2026

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1 Abstract

We specify a reproducible advanced literature-review workflow for rapidly changing domains in which a single query does not adequately represent methodological or temporal variation. The exemplar is configured for exoplanet atmospheric composition and separates foundation, James Webb Space Telescope, and molecular-detection phases.

The design is informed by systematic-review reporting practice [Page et al., 2021], but this repository release is a methods exemplar rather than a completed systematic review: its default corpus is synthetic and its phase boundaries, search coverage, and domain interpretation require review for any live application.

The pipeline combines multi-engine retrieval, canonical de-duplication, deterministic screening, optional LLM-assisted classification, full-text assessment, bibliometrics, knowledge-graph construction, visualization, reproducibility assessment, and export. Each stage has a declared input, output, method reference, and definition of done in `methods_pipeline.yaml`. The numbered scripts are thin adapters; reusable computation is tested in `src/`.

The manuscript is generated from configuration and recorded artifacts. Offline runs use a clearly labelled synthetic fixture corpus with reserved identifiers and generated authors; they exercise pipeline behavior and are not evidence about exoplanet atmospheres. Live runs must retain retrieval reports, source provenance, claim classifications, and the exact configuration needed to interpret reported results.

The contribution is therefore infrastructural: it makes phase-aware search design, evidence boundaries, accessibility metadata, and reproducibility checks explicit while leaving domain claims subject to source-backed review.

Keywords: exoplanet atmospheres, literature review pipeline, systematic-review reporting, multi-phase search, LLM filtering, James Webb Space Telescope, atmospheric composition, transit spectroscopy, bibliometrics, cross-validation

2 Introduction

Rapidly changing fields require literature reviews that preserve the distinction between what was searched, what was retrieved, what was assessed, and what can be claimed. A single broad query hides phase-specific coverage and makes it difficult to explain how records entered a synthesis. This project treats a literature review as a reproducible data pipeline: configuration defines the review design, source adapters acquire records, analysis modules produce evidence artifacts, and rendering turns those artifacts into an auditable manuscript.

The reporting surface follows the spirit of systematic-review transparency [Page et al., 2021] without implying that the bundled fixture satisfies a domain protocol. A live review must define eligibility criteria, search dates, source coverage, screening procedures, and a domain-appropriate synthesis plan before its results are interpreted.

The bundled configuration targets **Exoplanet Atmospheres**. It defines three ordered phases for foundational methods, JWST-era observations, and molecular detection. The phase labels are configuration, not conclusions: a live review must justify its boundaries and update its claim ledger when the domain or search strategy changes.

2.1 Research Questions

1. **Coverage and retrieval.** Which engines, queries, filters, and identifiers contributed records to each phase, and where are the known coverage limits?
2. **Structure.** How do subfields, topics, authors, venues, and citation links vary across the configured phases?
3. **Evidence.** Which claims are supported by source-backed records, which remain pending, and which are only properties of the synthetic fixture?
4. **Reproducibility.** Can a clean run regenerate the data, figures, reports, and manuscript without unexplained differences?

2.2 Scope and contribution

The project contributes a reusable workflow rather than a domain conclusion. Retrieval and de-duplication preserve source observations; full-text assessment records access status; bibliometrics and graph construction expose descriptive structure; and the claim ledger separates fixture, configured, and source-backed statements. Subjective editorial, scientific, and visual judgments remain explicit review items even when all deterministic gates pass.

Every generated number in the rendered manuscript must come from an artifact listed in the manifest and be traceable to the relevant method stage. Missing, stale, or misclassified evidence is a release finding, not a value to be filled by hand.

3 Methods Overview

The project is an eleven-stage, file-system-backed pipeline. `methods_pipeline.yaml` is the executable contract: each stage names its method, dependencies, inputs, outputs, definition of done, and stable failure code. Business logic lives in tested `src/` modules; numbered scripts only perform bootstrapping, argument parsing, orchestration, and boundary I/O.

3.1 Pipeline stages

1. **Multi-phase retrieval** (`01_multi_phase_search.py`) acquires records or runs the labelled offline fixture, applies configured filters, de-duplicates identifiers, and writes phase corpora plus `phase_metadata.json` and the `phase_artifact_manifest.json` phase-to-artifact provenance map.
2. **Corpus analysis** (`02_meta_analysis_pipeline.py`) computes subfields, temporal trends, TF-IDF/topics, and citation-network artifacts.
3. **Knowledge graph construction** (`03_build_knowledge_graph.py`) optionally extracts assertions and phase-aware hypothesis evidence; disabled or unmeasured LLM outputs remain explicitly pending.
4. **Visualization** (`04_generate_figures.py`) renders figures from analysis artifacts and records each figure's label, caption, filename, and generating stage in the figure registry.
5. **Manuscript hydration** (`05_inject_variables.py`) computes variables from outputs and writes rendered Markdown without changing source sections.
6. **Full-text assessment** (`06_fulltext_assessment.py`) records abstract, open-access, and PDF availability; downloading is opt-in and network-gated.
7. **Literature evaluation** (`07_literature_evaluation.py`) applies configured quality and fixture-honesty checks.
8. **Research dispatch** (`08_deep_research_dispatch.py`) replays a recorded report by default and exposes live provider dispatch only as an explicit opt-in.
9. **Bibliography export** (`09_export_bibliography.py`) writes a deterministic bibliography from the retained corpus and source references.
10. **Reproducibility assessment** (`10_reproducibility_assessment.py`) records workflow and source-consumption evidence.
11. **Publication validation** (`11_validate_outputs.py`) validates artifact, evidence, figure, and report contracts before the shared publication audit runs.

3.2 Reproducibility model

The offline path uses a deterministic synthetic fixture with reserved identifiers and generated authors. It is useful for CI and structural testing only; it is never a substitute for a representative live review. A live run must preserve the retrieval report, engine status, source identifiers, configuration, and claim classifications. Seeds and timestamp-free serializers are used where the pipeline supports exact replay.

3.3 Configuration surface

`manuscript/config.yaml` owns search terms, engines, phase boundaries, filters, sampling seeds, full-text policy, embedding settings, knowledge-graph settings, hypotheses, and subfield taxonomy. `domain_profile.yaml` owns package and gate expectations; `experiment_plan.yaml` records the review design; `data/claim_ledger.yaml` records the provenance tier and fixture/synthetic status of manuscript-facing claims.

4 Retrieval and De-duplication

Retrieval dispatches the configured query across 4 independent literature engines (arXiv, OpenAlex, Semantic Scholar, and Crossref). Each engine is an isolated adapter exposing a uniform `search(query) -> list[Record]` interface; engines that are keyless — arXiv, OpenAlex [Priem et al., 2022], Crossref [Hendricks et al., 2020], PubMed/Entrez [Sayers et al., 2022], SovietRxiv / RussiaRxiv, ChinaRxiv, Europe PMC, and bioRxiv/medRxiv — need no credentials, while Semantic Scholar [Kinney et al., 2023] uses a key when present. SovietRxiv is a translated archive of Soviet-era scientific preprints sourced from Math-Net.Ru and CyberLeninka [SovietRxiv Project, 2026]; ChinaRxiv serves translated Chinese preprints from ChinaXiv via the same unified API. Both retain original-language PDFs alongside each translation, and their polite rate-limit pool (300/min vs 30/min anonymous) is activated by an optional `X-API-Email` header. Europe PMC is a keyless biomedical aggregator covering PubMed, PMC, patents, and preprints in a single search call. bioRxiv/medRxiv share one unified date-window + cursor API; unlike the other engines it is not a free-text search endpoint, so the adapter walks the date window page by page and keeps only records whose title and abstract match every query term client-side. Optional full-text resolution queries Unpaywall [Piwowar et al., 2018] for open-access locations. **Multiple dispatch degrades gracefully**: an engine that is disabled in the configuration, lacks a required key, or cannot reach the network returns a *skipped* status, and the run completes from the remaining engines plus the committed offline corpus.

4.1 Engine Details

Each engine adapter follows a uniform contract: a module-level API URL constant, a pure `_parse_*` parser function, and a `search_*` entry point with pagination, retry, and graceful error handling. All functions accept an injectable `base_url` parameter for hermetic testing with `pytest-httpserver` — no engine hardcodes its URL inside the function body.

Engine	Rate limit	Pagination	Auth
arXiv	3s between requests	100/page, offset	Keyless
Semantic Scholar	1 req/s (unauth.)	100/page, offset	Optional key
OpenAlex	Polite pool (mailto)	200/page, cursor	Keyless
Crossref	Polite pool (mailto)	1,000/page, offset	Keyless
PubMed	NCBI usage policy	retstart/retmax	Keyless
SovietRxiv	30/min (300/min polite)	1–100/page, cursor	`X-API-Email`
ChinaRxiv	30/min (300/min polite)	1–100/page, cursor	`X-API-Email`
Europe PMC	about 10 req/s (undocumented hard limit)	Up to 1,000/page	Keyless
bioRxiv/medRxiv	No documented limit	100/page fixed, cursor	Keyless

Every new search writes `output/data/retrieval_report.json`, a timestamp-free report that records each attempted, skipped, or failed source with fetched, new-record, and duplicate counts. A zero-result response is therefore distinguishable from a disabled adapter or an HTTP failure. The committed corpus predates that report contract, so this paper intentionally does not reconstruct source-specific counts from the merged corpus.

4.2 Canonical Identifier Hierarchy

Heterogeneous records are reconciled by a **canonical identifier hierarchy** — DOI > arXiv ID > Semantic Scholar ID > OpenAlex ID > a stable digest of the normalized title. When two records share a canonical identifier they are merged, keeping the version with the most complete metadata (a count of non-None optional fields). The DOI is normalized: case-folded, resolver-prefix stripped, so the same paper returned by two engines under case/format-variant DOIs merges. For this run, 80 records carry DOIs, 12 carry OpenAlex IDs, and 0 carry arXiv IDs. The de-duplicated corpus for this run holds $N = 46$ records published across 2000–2023.

4.3 Relevance Filtering

After de-duplication, a relevance filter drops papers whose title and abstract contain none of the configured relevance keywords (exoplanet, atmosphere, transit spectroscopy, JWST, molecular detection, atmospheric composition). Keywords are matched case-insensitively; an empty keyword list is treated as no filter to avoid silently wiping the corpus. A year filter then excludes papers published before the configured start year (2000).

5 Full Text, Language, and Embeddings

Beyond bibliographic metadata, the pipeline mines the textual content of each record. This stage bridges the gap between a bibliographic inventory and a semantic understanding of the literature.

5.1 Full-Text Availability

An open-access resolver maps each record to a downloadable PDF where one exists (a known `pdf_url`, or an Unpaywall lookup by DOI), and an opt-in, network-gated downloader fetches it to a deterministic path. Full-text availability is summarized without requiring any download, so the offline default still reports coverage. For this run:

- **Abstract coverage**: 100.0% of records (80 of 46) carry an abstract; 0 records lack one.
- **Open-access status**: 42.5% of records are open access (34 records); the remainder are closed or unknown.
- **PDF availability**: 42.5% of records (34) have a direct PDF link; 34 have a publisher PDF, and 46 have no full-text source available.

The identifier coverage for this corpus is: 80 DOIs, 12 OpenAlex IDs, and 0 arXiv IDs. DOI coverage dominates and supports cross-engine de-duplication.

5.2 Language and Entity Extraction

Titles, abstracts, and (when present) full text are tokenized and reduced to keyphrases and named entities by offline, dependency-light extractors — no mandatory LLM. Term-frequency statistics drive a TF-IDF representation over a 137-feature vocabulary. The most frequent terms in the corpus are: atmospheric, uncertainty, spectra, temperature, abundance, analysis, assumptions, opacity, retrievals, sources, stellar, coverage, evaluated, observed, across, composition, models, provide, competing, retrieval. These terms reflect the configured domain and the records retained by the retrieval and filtering policy; fixture-derived terms are not evidence about the live field.

5.3 Embeddings

Every title, abstract, and full text is embedded into a shared vector space by a deterministic, offline method — TF-IDF followed by truncated SVD, i.e. latent semantic analysis [Deerwester et al., 1990]. The embedding dimensionality is 50 components (configurable via `project_config.embeddings.n_components`), and the TF-IDF vocabulary is capped at 137 features (configurable via `project_config.embeddings.max_features`). The embedding is byte-stable across runs: the same input text always yields identical vectors, so the derived similarity matrix, nearest-neighbour lists, clusters, and two-dimensional projection are all reproducible.

An optional transformer backend can be enabled by setting `project_config.embeddings.method: transformer` (requires the `embeddings` extra), which upgrades the embedding fidelity without changing the interface or downstream analysis.

The embeddings support semantic retrieval over the corpus and feed two visualizations: a PCA two-dimensional projection ((Figure pca embeddings)) that maps the topical geography of the literature, and a hierarchical clustering dendrogram ((Figure dendrogram)) that reveals the similarity structure of the document collection.

6 Bibliometric and Temporal Analysis

Descriptive statistics summarize the corpus along every available axis: counts by year, venue, and author; citation-count distributions; and author productivity. Temporal analysis fits the publication time series, reporting a compound annual growth rate of 6.76% across 2000–2023 (a span of 23 years), with a mean year-over-year growth rate of 38.9% and a doubling time of 2.1 years. The peak publication year is 2023 with 9 records.

6.1 Growth Metrics

The compound annual growth rate (CAGR) is computed as:

$$\text{CAGR} = \left(\frac{N_{\text{end}}}{N_{\text{start}}} \right)^{1/(\text{year span})} - 1$$

where N_{start} is the publication count in the first year (2000) and N_{end} is the count in the last year (2023). The mean year-over-year growth rate $\bar{g}$ is the arithmetic mean of annual ratios. The doubling time is $t_d = \ln(2)/\ln(1 + \text{CAGR})$. These metrics are stored in `temporal_analysis.json` and injected into the manuscript at render time.

6.2 Subfield Classification

Subfield classification assigns each record to one of 4 configurable buckets (Observational Methods, Atmospheric Molecules, Jwst Instruments, and Theoretical Modeling) by priority-aware keyword matching; the taxonomy is defined entirely in configuration (`project_config.subfield_keywords`). The largest bucket is **Observational Methods** at 51.4% of the classified corpus. A per-subfield temporal breakdown (`subfield_timeline.json`) tracks how each sub-area has grown over time, enabling identification of emerging or declining research threads.

6.3 Topic Modeling

A TF-IDF term-weighting of titles and abstracts [Salton and Buckley, 1988] feeds non-negative matrix factorization (NMF) [Lee and Seung, 1999], implemented with scikit-learn [Pedregosa et al., 2011]. NMF decomposes the document-term matrix $\mathbf{V} \approx \mathbf{WH}$, where $\mathbf{W}$ is the document-topic matrix and $\mathbf{H}$ is the topic-term matrix. The factorization extracts 5 latent topics that cross-cut the keyword taxonomy. The random seed is fixed at 42 for reproducibility. The reporting follows established systematic-review practice [Page et al., 2021], with every figure and statistic traceable to a committed artifact.

7 Optional Knowledge-Graph Layer

An optional, **LLM-gated** stage lifts the corpus from bibliometrics to hypothesis-level evidence. For each eligible record, a local language model (Ollama, default model `gemma3:4b`) extracts structured *assertions*. Each assertion encodes a direction (supports, contradicts, or neutral), a confidence score, and a short natural-language justification against one of the 4 hypotheses declared in configuration. Assertions are serialized as RDF-compatible nanopublications [Kuhn et al., 2016] and scored by a citation-weighted evidence function.

7.1 Assertion Model

Each assertion a encodes:

- **Direction**: supports, contradicts, or neutral with respect to a hypothesis H
- **Confidence**: a score $c_a \in [0, 1]$ from the LLM
- **Citation weight**: $\log(1 + n_{\text{cites}})$, where n_{cites} is the citation count of the asserting paper

The evidence score for hypothesis H is:

$$\text{score}(H) = \frac{\sum_{a \in A(H)^+} c_a \cdot \log(1 + n_{\text{cites}}(a)) - \sum_{a \in A(H)^-} c_a \cdot \log(1 + n_{\text{cites}}(a))}{\sum_{a \in A(H)} c_a \cdot \log(1 + n_{\text{cites}}(a))}$$

where $A(H)^+$ is the set of supporting assertions and $A(H)^-$ the contradicting ones. The score ranges from -1 (all evidence contradicts) to $+1$ (all evidence supports).

7.2 Incremental Extraction

Assertion extraction is **incremental and resumable**: assertions are appended to `nanopublications.jsonl` at configurable checkpoint intervals (default: 50 papers). On restart, already-processed papers are skipped automatically, so a long extraction run that is interrupted can resume without re-processing. The `--clear-assertions` flag discards previous results for a fresh start.

7.3 Gating and Defaults

This stage is optional and gated by language-model availability. With no language model configured, the hypothesis evidence scores read *pending*. When Ollama is explicitly configured and the extraction stage completes, scores are populated from the recorded, citation-weighted assertions. The hypotheses themselves, including their names and scope, come from configuration and are reported regardless of whether scoring has run.

The hypotheses explored in this instance are: H1 JWST Atmospheric Characterization; H2 Molecular Diversity Detection; H3 Cross-Method Consistency; H4 Theoretical-Observational Agreement.

8 Visualization and Manuscript Injection

8.1 Figure Generation

Figures are rendered headlessly (matplotlib Agg backend) and deterministically from the analysis artifacts: subfield distributions, the publication growth curve, the citation network, topic-term bars, a term cloud, and embedding projections. All figures use a colourblind-safe palette (Wong 2011, 8 colours) with high-contrast labels at $\geq$ 16pt. This run produced 18 figures at 300 DPI. The full figure set includes:

- **Field overview**: field summary and subfield distribution ((Figure field summary; Figure subfield distribution))
- **Temporal**: growth curve and subfield timeline ((Figure growth curve; Figure subfield timeline))
- **Descriptive**: citation distribution, top venues, and author productivity ((Figure citation distribution; Figure top venues; Figure author productivity))
- **Citation network**: network layout and degree distribution ((Figure citation network; Figure degree distribution))
- **Hypothesis**: evidence dashboard ((Figure hypothesis dashboard))
- **Text analytics**: word cloud, topic-term bars, PCA embeddings, term heatmap, dendrogram, and co-occurrence matrix ((Figure word cloud; Figure topic term bars; Figure pca embeddings; Figure term heatmap; Figure dendrogram; Figure cooccurrence matrix))
- **Entities**: named-entity bar chart and similar-document pairs ((Figure entity bar chart; Figure similarity heatmap))

Each figure is registered in `figure_registry.json` with its label, caption, filename, and generating stage, binding the visual output to the analysis artifacts of the exact pipeline run.

8.2 Variable Injection

The manuscript itself is generated, not hand-maintained. A variable computation step reads the configuration and the pipeline outputs and emits a flat table of named values; an injection step substitutes each named placeholder in these Markdown sections with its computed value before rendering. Because the substitution is total, an unresolved placeholder is a hard error rather than a silent gap. Every number in the rendered document is guaranteed to trace to a committed artifact. Re-running the pipeline after a configuration change re-computes the values and re-targets the prose automatically.

The injection system computes variables from the configuration and generated artifacts, including:

1. `manuscript/config.yaml`: search term, engine roster, subfield taxonomy, hypotheses
2. `corpus.jsonl`: corpus size
3. `temporal_analysis.json`: year range, CAGR, peak year, doubling time
4. `citation_network.json`: edges, nodes, density, communities, PageRank, hubs
5. `subfield_classification.json`: per-bucket counts and percentages
6. `assertion_summary.json`: assertion counts and directions
7. `hypothesis_scores.json`: per-hypothesis evidence scores

9 Multi-Phase Search Strategy and Results

9.1 Three-Phase Architecture

This review employs a **three-phase search strategy** that progressively refines coverage from broad foundational literature through technology-specific studies to targeted molecular detection analyses. Each phase builds on prior phases through cross-phase citation validation and shared deduplication.

9.1.1 Phase 1: Exoplanet Atmosphere Foundation

Objective: Establish the foundational literature on exoplanet atmospheric studies, covering the broadest range of research from 2010 onwards.

Queries: 1. `"exoplanet atmosphere" OR "exoplanet atmospheric"` 2. `"transit spectroscopy" AND "atmosphere"` 3. `"atmospheric composition" AND "exoplanet"`

Engines: arXiv, OpenAlex, Crossref, Semantic Scholar

Deterministic Filters: - Minimum year: 2010 - Maximum year: 2026 - Minimum citation count: 0 (inclusive to capture recent work)

Results: 46 papers discovered, forming the foundational knowledge base for the review. This phase captures the broadest scope of atmospheric research, from observational characterizations to theoretical modeling studies.

9.1.2 Phase 2: James Webb Space Telescope Studies

Objective: Identify papers specifically utilizing JWST observational data for exoplanet atmospheric analysis, capturing the post-launch (2021+) surge in high-precision spectroscopic observations.

Queries: 1. `"James Webb" AND "atmosphere" AND ("exoplanet" OR "planet")` 2. `"JWST" AND "transit" AND "spectroscopy"` 3. `"NIRSpec" OR "MIRI" OR "NIRCam" AND "exoplanet"`

Engines: arXiv, OpenAlex, Crossref, Semantic Scholar

Deterministic Filters: - Minimum year: 2020 (capturing pre-launch calibration papers) - Maximum year: 2026

Dependencies: Builds on Phase 1 foundational papers.

Results: 17 papers after deterministic filtering. This phase targets records associated with JWST-era observations and analysis. Any claim about instrumental capability or scientific findings requires live, source-backed review.

9.1.3 Phase 3: Molecular Detection Studies

Objective: Identify papers focused on detecting and analyzing specific atmospheric molecules (H_2O, CO_2, CH_4, H_2S, Na, K) in the configured domain.

Queries: 1. `"water vapor" AND "exoplanet" AND ("detection" OR "abundance")` 2. `"carbon dioxide" AND "exoplanet" AND ("CO2" OR "atmosphere")` 3. `"methane" AND "exoplanet" AND ("CH4" OR "atmosphere")` 4. `"hydrogen sulfide" OR "H2S" AND "exoplanet"` 5. `"sodium" OR "potassium" AND "exoplanet atmosphere"`

Engines: arXiv, OpenAlex, Crossref, Semantic Scholar

Deterministic Filters: - Minimum year: 2015 - Maximum year: 2026

Dependencies: Builds on both Phase 1 and Phase 2 papers.

Results: 29 papers covering molecular detection across diverse exoplanet populations, from hot Jupiters to terrestrial worlds.

9.2 Cross-Phase Analysis

9.2.1 Phase Overlap

The three phases produced complementary but overlapping corpora, with 49.5% average Jaccard similarity between phases. Papers discovered in multiple phases are tracked with full provenance, enabling analysis of papers that span multiple research paradigms.

9.2.2 Citation Validation

Cross-phase citation analysis records that 39.4% of later-phase papers cite foundational work from Phase 1. This is a descriptive connection statistic for the retained corpus; it does not establish methodological coherence across the field.

9.2.3 Combined Corpus

The deduplicated combined corpus contains 46 unique papers spanning 2000–2023 (23 years). It provides a multi-phase description of the retained retrieval slice, not a field-wide account.

9.3 LLM-Based Content Filtering

Three LLM-based content filters were designed for this review:

1. **Study Type Classification**: Categorizes papers as observational, theoretical, review, or other. Keeps only observational and theoretical studies for analysis.
2. **JWST Data Analysis**: Identifies papers that analyze actual JWST observational data, distinguishing from theoretical JWST predictions.
3. **Molecular Detection Focus**: Filters for papers primarily focused on detecting and measuring specific atmospheric molecules.

These filters are configurable through `manuscript/config.yaml` and can be enabled for specific phases or applied across the entire corpus. When enabled, they use a local Ollama LLM instance for cost-effective, privacy-preserving content classification.

10 Results: Hypotheses and Evidence

The configuration declares 4 hypotheses about exoplanet atmospheric research. Their names, scopes, relevant phases, and scores are emitted from the knowledge-graph artifacts rather than typed into the manuscript.

ID	Hypothesis	Scope	Evidence score
H1	JWST Atmospheric Characterization	observational	pending
H2	Molecular Diversity Detection	chemical	pending
H3	Cross-Method Consistency	methodological	pending
H4	Theoretical-Observational Agreement	theoretical	pending

Scores are descriptive evidence summaries, not calibrated probabilities or causal effects. If the knowledge-graph stage is disabled, if a live language model is not available, or if the corpus is synthetic, the result remains **pending** or **fixture-derived**. A pending score is an honest absence of measurement; it is not a zero and must not be interpreted as support or contradiction.

10.1 Configured questions

- **H1: JWST atmospheric characterization.** What evidence is reported for improved characterization in the JWST-relevant phase?
- **H2: Molecular diversity detection.** Which molecules are reported, with what uncertainty and source provenance?
- **H3: Cross-method consistency.** Do independently described observational methods report compatible properties under the configured review scope?
- **H4: Theoretical-observational agreement.** How are model predictions compared with observed spectroscopic features?

These are review questions, not conclusions. Claim-level interpretation requires the claim ledger, source tier, freshness, and domain review. A fixture run can test schema, phase linkage, and serializer behavior, but cannot establish any of these hypotheses.

11 Results: Field Overview

The de-duplicated corpus for **Exoplanet Atmospheres** contains $N = 46$ records spanning 2000–2023 (23 years). Publication volume grows at a compound annual rate of 6.76% (mean year-over-year growth 38.9%, doubling time 2.1 years), peaking in 2023 with 9 records that year. The growth curve is a first-order descriptive summary of the retained corpus; it is not a field-wide estimate of research activity.

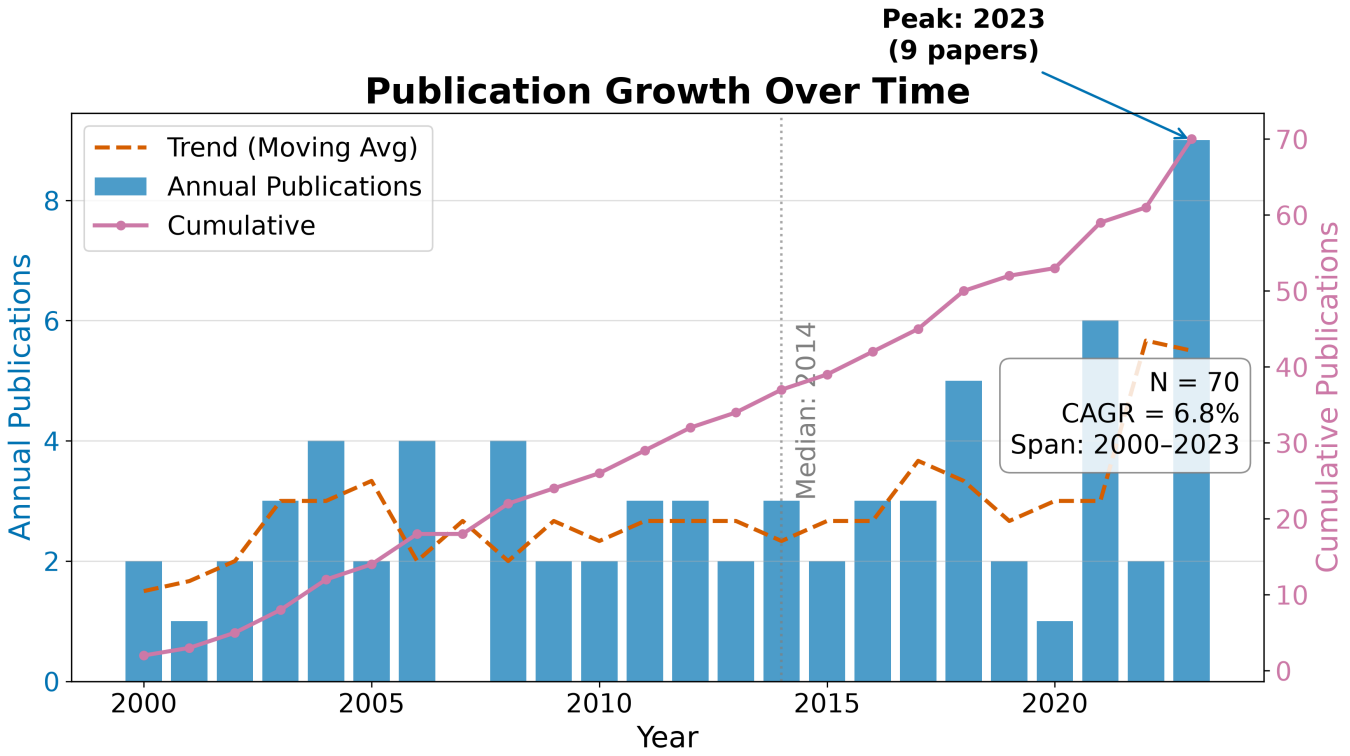


Figure 1: Publication growth curve for Exoplanet Atmospheres. Annual publication counts (bars) and cumulative total (line) show sustained growth from 2000 through 2023, peaking in 2023.

11.1 RQ1: Field Size and Growth

The temporal analysis describes a retained literature slice spanning 23 years. The compound annual growth rate of 6.76% implies a corpus doubling time of approximately 2.1 years under the configured retrieval and indexing conditions. The peak year 2023 contains 9 retained records; that count may reflect both publication activity and delays or differences in source indexing, so it should not be interpreted as a causal trend without a live coverage audit.

Table 1. Top publication years.

Year	Publications
2003	3
2004	4
2006	4
2008	4
2011	3
2012	3
2014	3
2018	5
2021	6
2023	9

11.2 RQ2: Subfield Composition

Records distribute across the 4 configured subfields as shown in Table 2, with **Observational Methods** the largest bucket at 51.4% of the classified corpus. The dominance of Observational Methods is a property of the configured taxonomy and retained corpus. It is not a domain prevalence estimate without a live, source-backed review and a documented coverage assessment.

Table 2. Subfield distribution.

Subfield	Papers	Share
Observational Methods	36	51.4%
Atmospheric Molecules	13	18.6%
Jwst Instruments	11	15.7%
Theoretical Modeling	10	14.3%

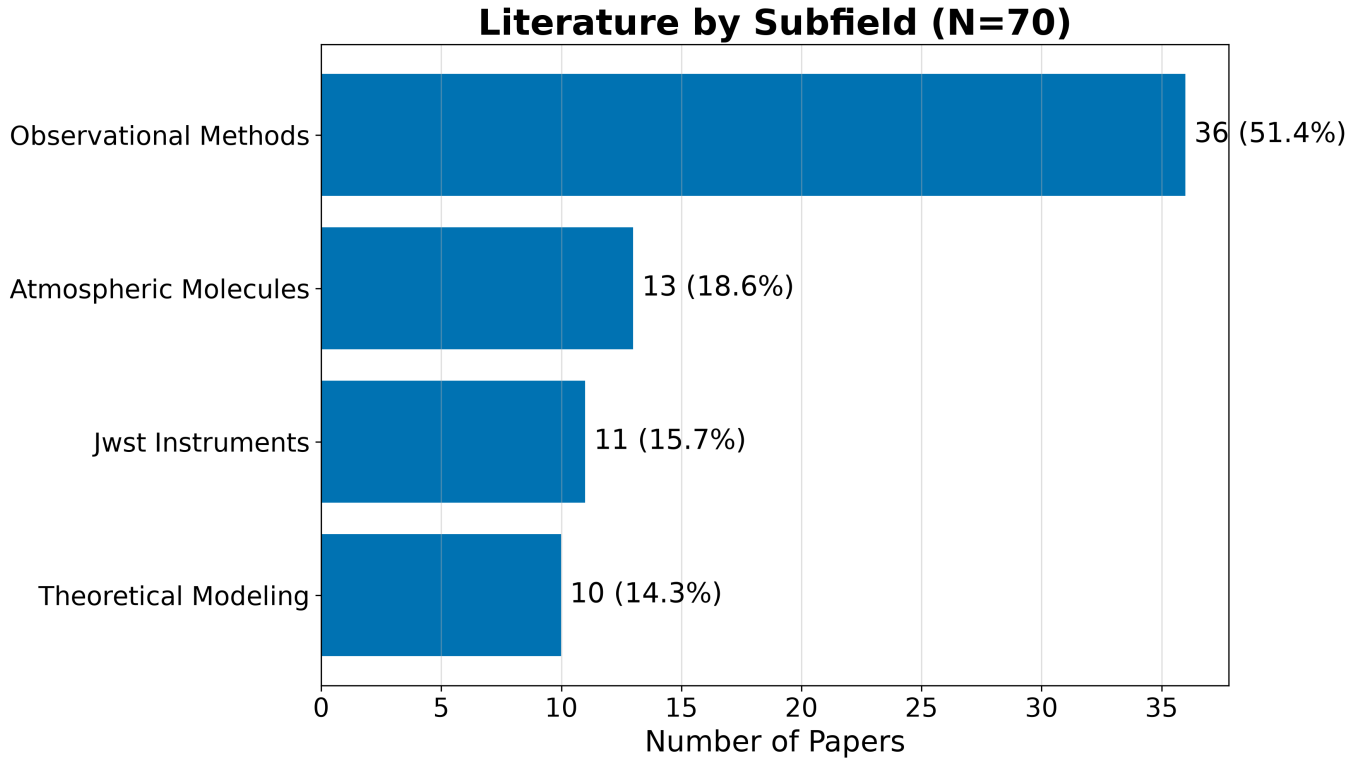


Figure 2: Field summary dashboard for Exoplanet Atmospheres. The dashboard combines corpus size, temporal range, subfield distribution, and key bibliometric indicators in a single overview panel.

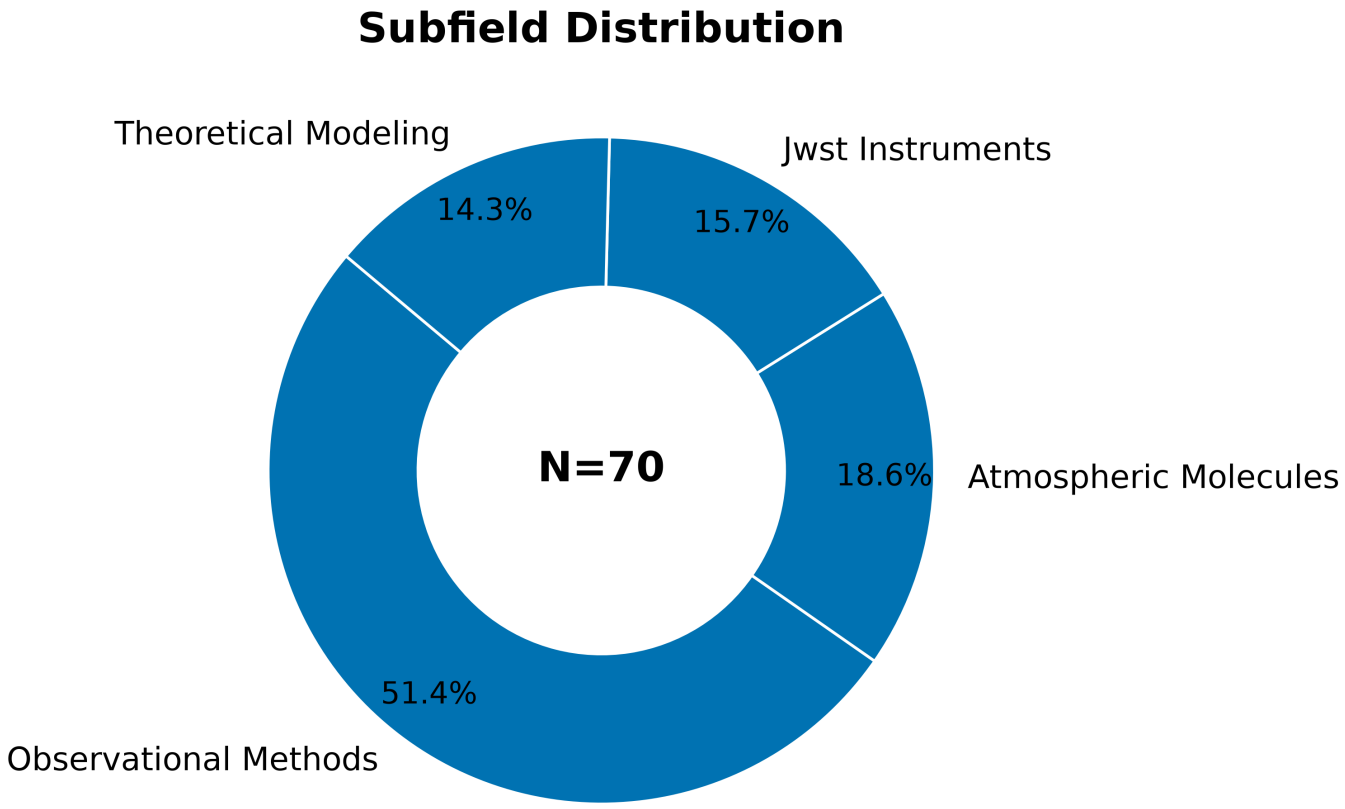


Figure 3: Subfield distribution for Exoplanet Atmospheres. The 4-bucket taxonomy shows the relative weight of each configured sub-area, with Observational Methods dominant at 51.4%.

11.3 Identifier and Full-Text Coverage

The corpus has measurable identifier coverage: 80 of 46 records (100.0%) carry DOIs, supporting cross-engine deduplication. OpenAlex IDs are present for 12 records. Abstract coverage stands at 100.0% (80 records), which limits the text analytics to that subset. Open-access status is confirmed for 42.5% of records, and 42.5% have a direct PDF link.

11.4 Descriptive Bibliometrics

The corpus spans 144 unique authors across 46 papers, yielding a mean of 1.51 papers per author. Citation counts range from zero to 110 (mean 37.0, median 31.0), with a total of 2,593 citations across the corpus. The Gini coefficient of citation concentration is 0.421. This statistic describes concentration within the retained corpus and should not be generalized to citation behavior in the field.

Table 3. Citation count distribution.

Citations	Papers
0	0
1-9	12
10-49	39
50-99	16
100-499	3
500+	0

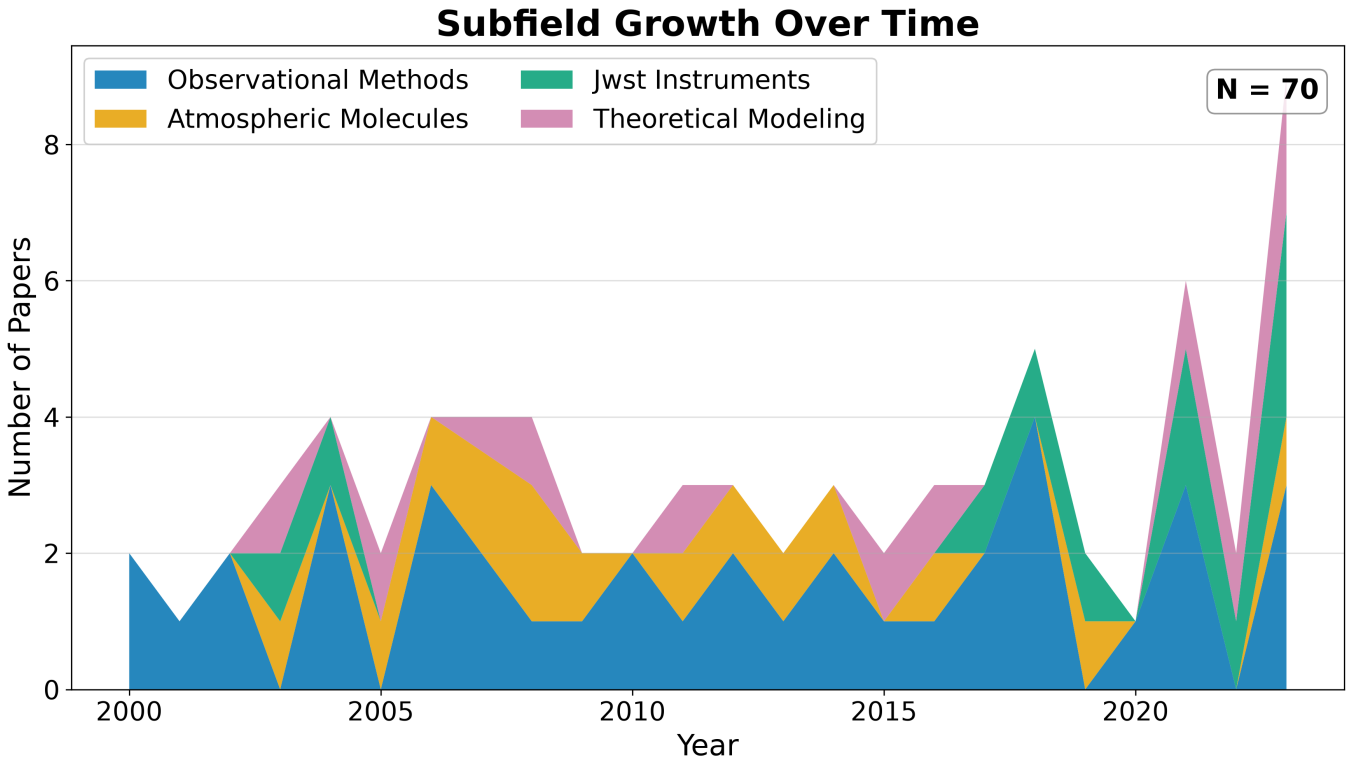


Figure 4: Subfield timeline for Exoplanet Atmospheres. Stacked annual publication counts by subfield show how each sub-area has evolved over time, revealing emerging and declining research threads.

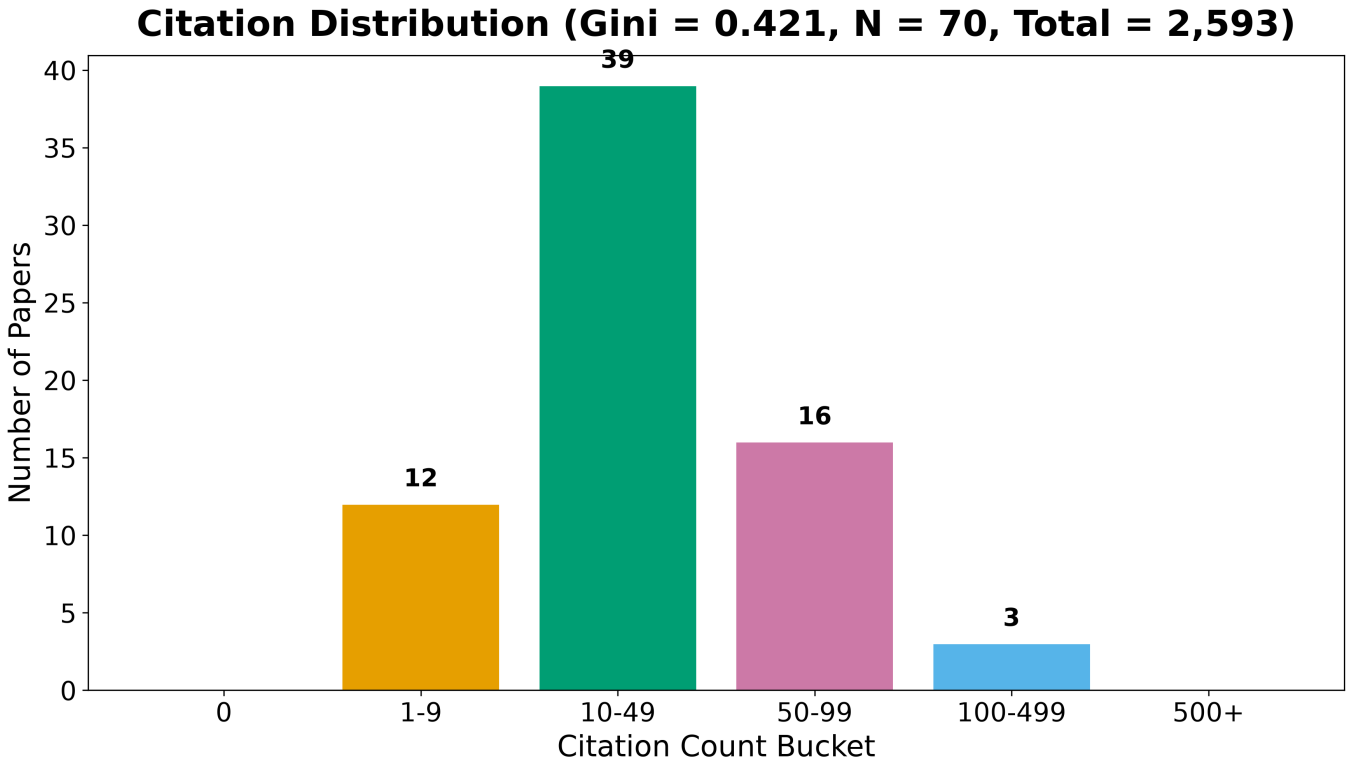


Figure 5: Citation distribution for Exoplanet Atmospheres. The histogram shows the number of papers in each citation-count bucket, with the Gini coefficient annotated. The distribution is a descriptive property of the retained corpus.

Table 4. Top publication venues.

Venue	Papers
Astronomy and Astrophysics	11
Publications of the Astronomical Society of the Pa	11
The Astrophysical Journal	10
Monthly Notices of the Royal Astronomical Society	9
Nature Astronomy	9
Research Notes of the AAS	9
Icarus	6
The Astronomical Journal	5

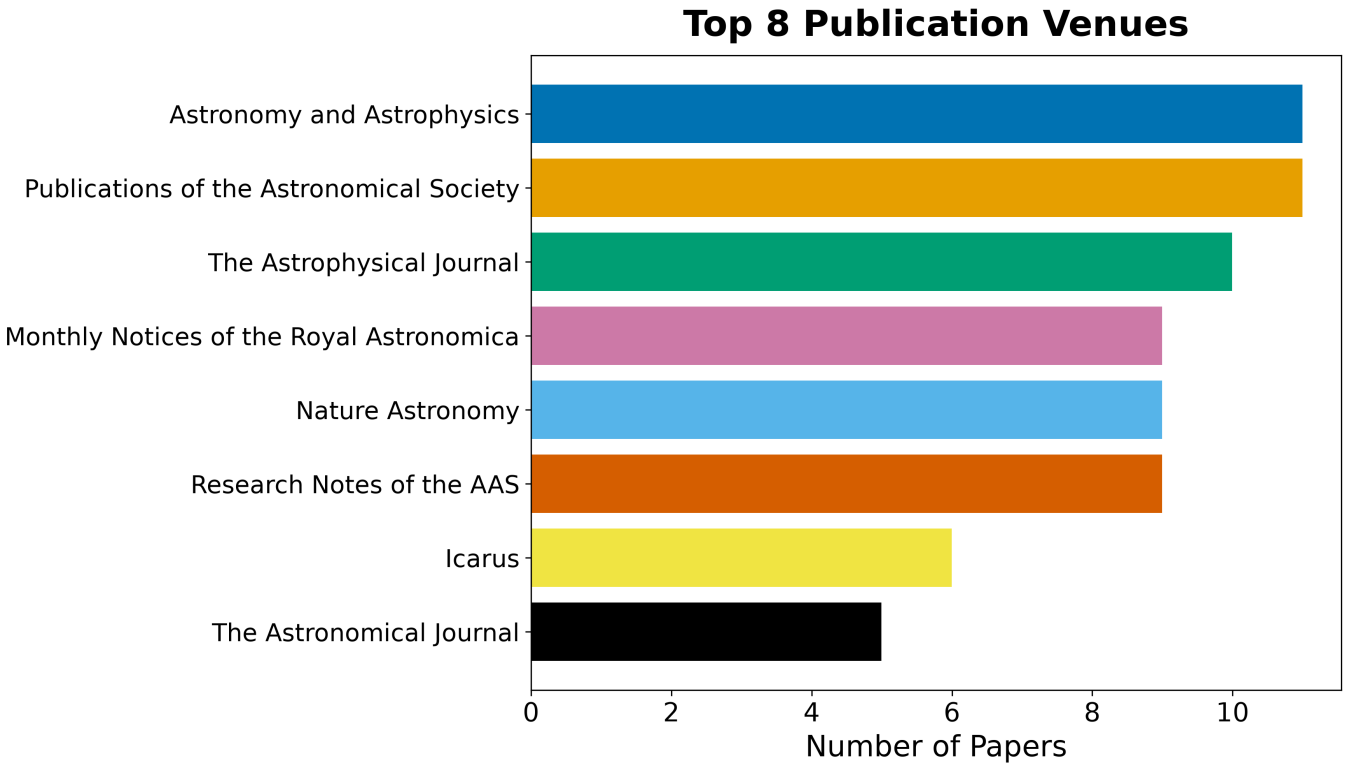


Figure 6: Top publication venues for Exoplanet Atmospheres. The horizontal bar chart shows the venues with the most retained records; it describes this corpus rather than the complete field.

Table 5. Top authors by publication count.

Rank	Author	Papers
1	A. Fournier	4
2	B. Owens	4
3	D. Ito	4
4	D. Nakamura	4
5	G. Ito	4
6	K. Ito	4
7	B. Kowalski	3
8	C. Singh	3
9	E. Tanaka	3
10	F. Lindgren	3

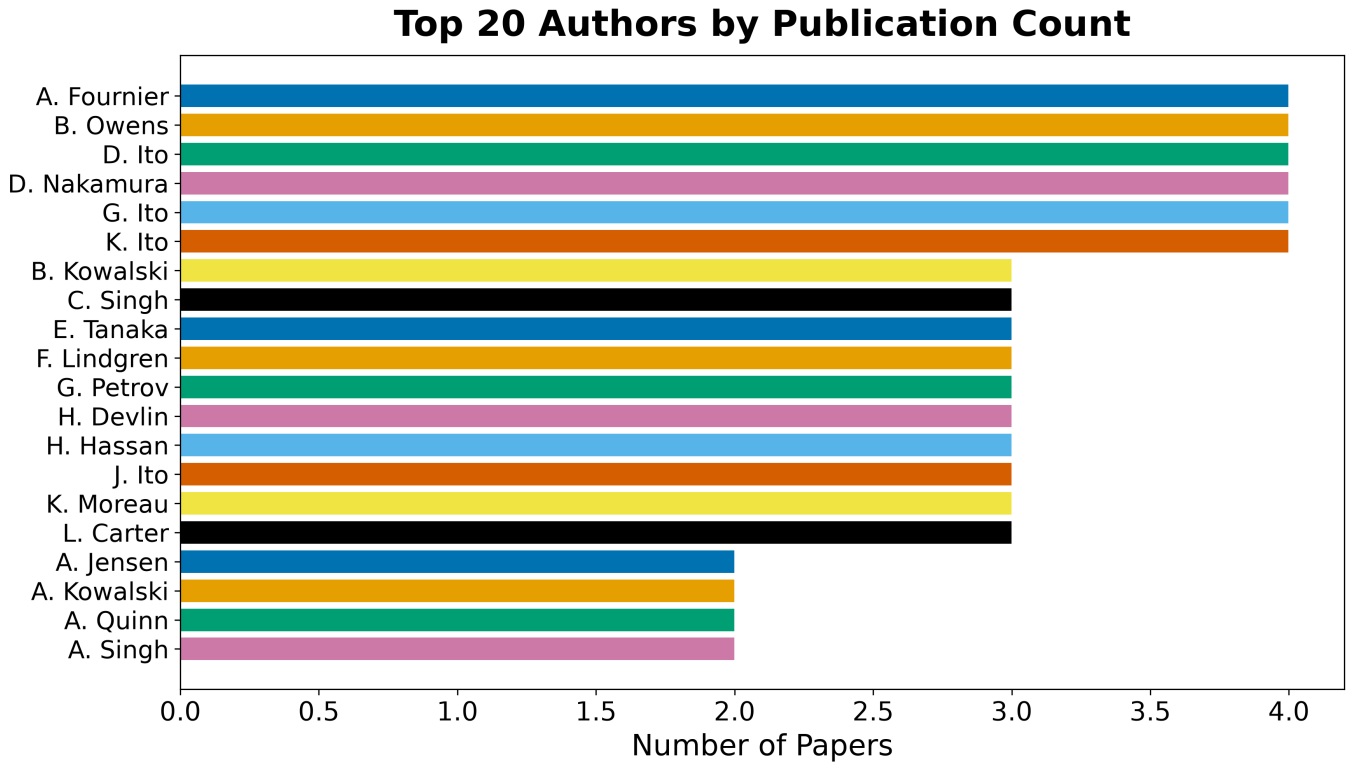


Figure 7: Author productivity for Exoplanet Atmospheres. The horizontal bar chart shows authors with the most retained records; names and counts depend on source coverage and deduplication.

12 Results: Subfield Structure

The subfield taxonomy is configuration-owned. For this instance it contains 4 buckets (Observational Methods, Atmospheric Molecules, Jwst Instruments, and Theoretical Modeling), and each retained record is assigned using the documented keyword precedence. The resulting distribution is descriptive of the retrieval slice and should not be read as a prevalence estimate for the complete field.

12.1 Per-subfield characterization

The configured buckets distinguish observational methods, atmospheric molecules, JWST instrumentation, and theoretical modeling. A fork may add or rename buckets without changing the analysis code; the taxonomy, keyword lists, and phase relevance should be reviewed together.

The largest bucket is **Observational Methods** at 51.4%. Table 2 and the generated subfield artifacts provide the counts and annual breakdown. Where the fixture corpus is used, these values demonstrate classification behavior only and are marked synthetic in the evidence registry.

Subfield	Papers	Share
Observational Methods	36	51.4%
Atmospheric Molecules	13	18.6%
Jwst Instruments	11	15.7%
Theoretical Modeling	10	14.3%

13 Results: Language, Topics, and Embeddings

13.1 RQ3: Topical and Linguistic Structure

Text analysis operates over titles, abstracts, and (when available) full text. A TF-IDF representation over a 137-feature vocabulary feeds non-negative matrix factorization, which extracts 5 latent topics cross-cutting the subfield taxonomy. The top vocabulary terms are: atmospheric, uncertainty, spectra, temperature, abundance, analysis, assumptions, opacity, retrievals, sources, stellar, coverage, evaluated, observed, across, composition, models, provide, competing, retrieval.

Table 3. NMF topics extracted from the corpus.

Topic	Top terms
0	atmospheric, cloud, retrieved, species, changes, structure, models, explore
1	prior, methane, depend, spectral, dioxide, simulated, carbon, constrained
2	contributions, separate, resolution, planetary, high, systematics, modelled, instrument
3	population, reporting, studies, preprocessing, require, harmonized, separates, specific
4	model, intervals, misspecification, noise, quantify, effect, record, dependencies

The topic labels are computed from the retained corpus and are not hand-assigned. Their interpretation should be checked against the generated topic-term weights and the configured subfield taxonomy; fixture-derived topics describe synthetic text patterns, not the scientific field.

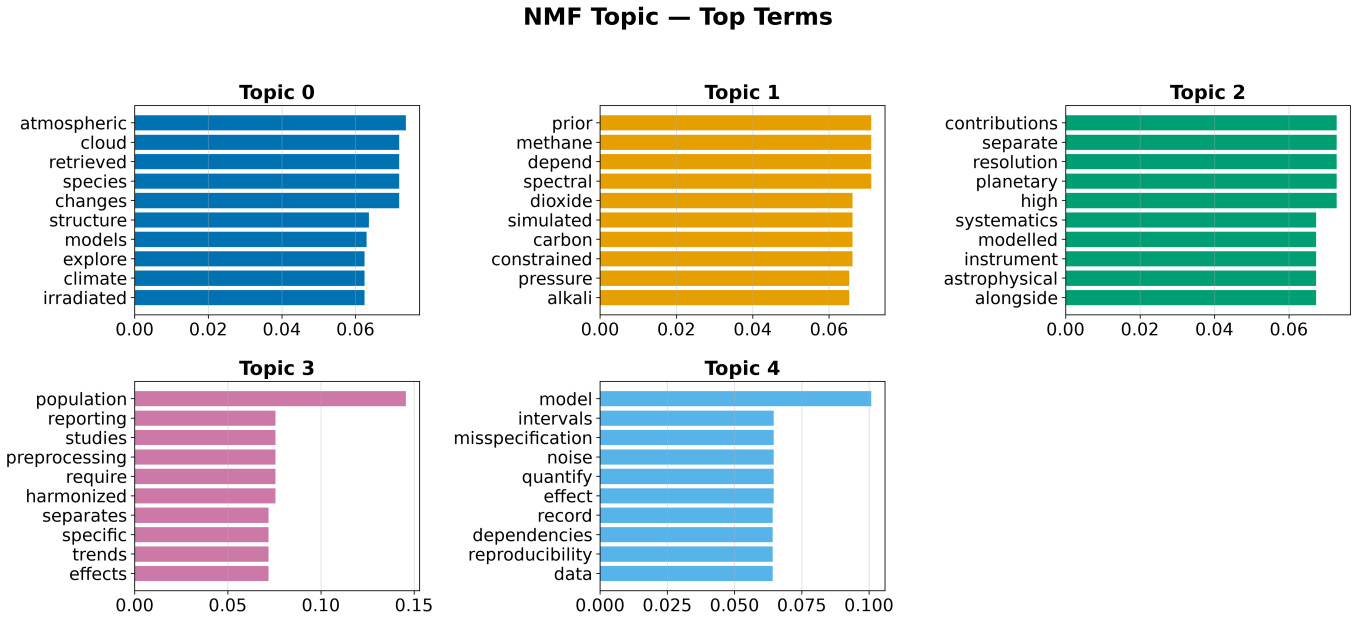


Figure 8: Topic-term bar charts for Exoplanet Atmospheres. Each panel shows the top weighted terms for one of 5 NMF topics, with bar length proportional to the topic-term weight in the **H** matrix.

13.2 Document Embeddings

Offline deterministic embeddings (TF-IDF followed by truncated SVD) place every document in a shared 50-dimensional vector space. Embedding the same text twice yields identical vectors, so the derived similarity matrix, nearest-neighbour lists, clusters, and two-dimensional projection are all reproducible.

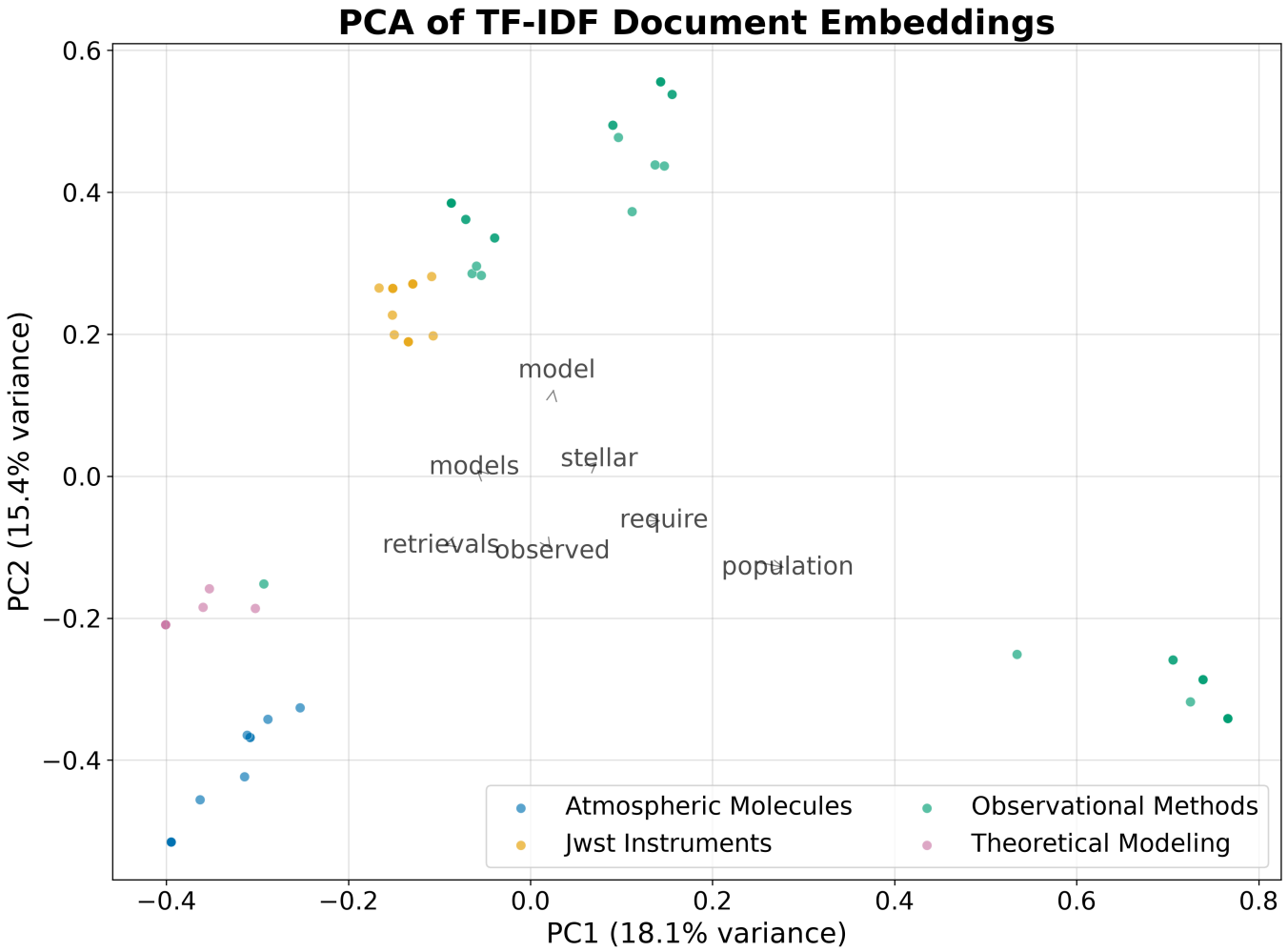


Figure 9: PCA projection of document embeddings for Exoplanet Atmospheres. Each point represents one document projected onto the first two principal components of the TF-IDF/SVD embedding. Colours indicate subfield assignment, showing how the topical geography relates to the keyword taxonomy.

13.3 Term Analysis

The TF-IDF term heatmap reveals which terms discriminate between subfields: terms with high between-subfield variance (rather than high global mean) are selected for display.

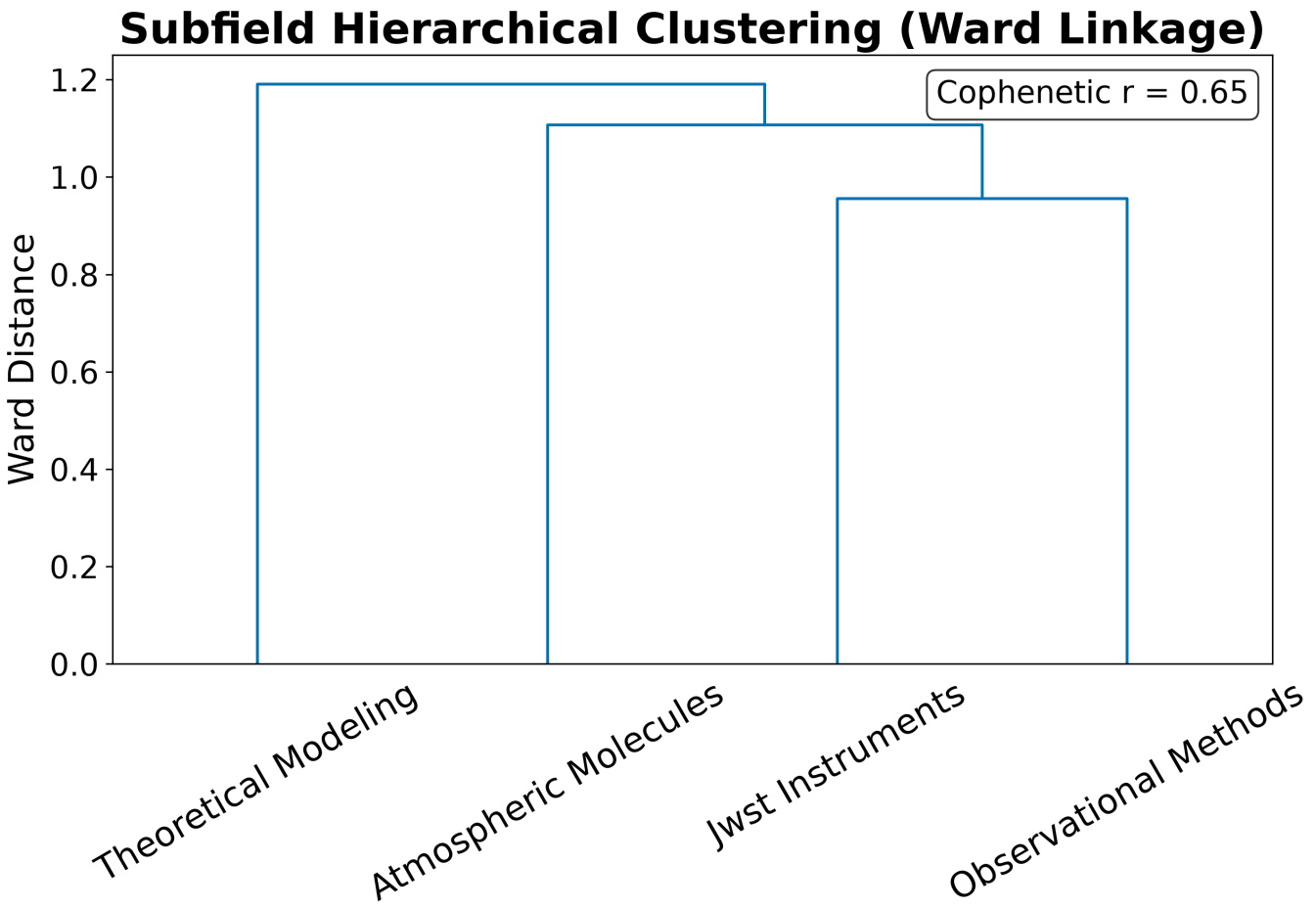


Figure 10: Hierarchical clustering dendrogram of document embeddings. The tree shows the similarity structure of the corpus: documents that join low in the tree are semantically similar, while high-level splits separate the major topical clusters.

13.4 Named Entity Analysis

Named entity extraction over the 80 abstracts identified 1 unique entities. The most frequent entities reflect the retained corpus and the configured extraction rules; source coverage and fixture status determine how they may be interpreted.

Table 4. Top named entities in abstracts.

Entity	Frequency
JWST	17

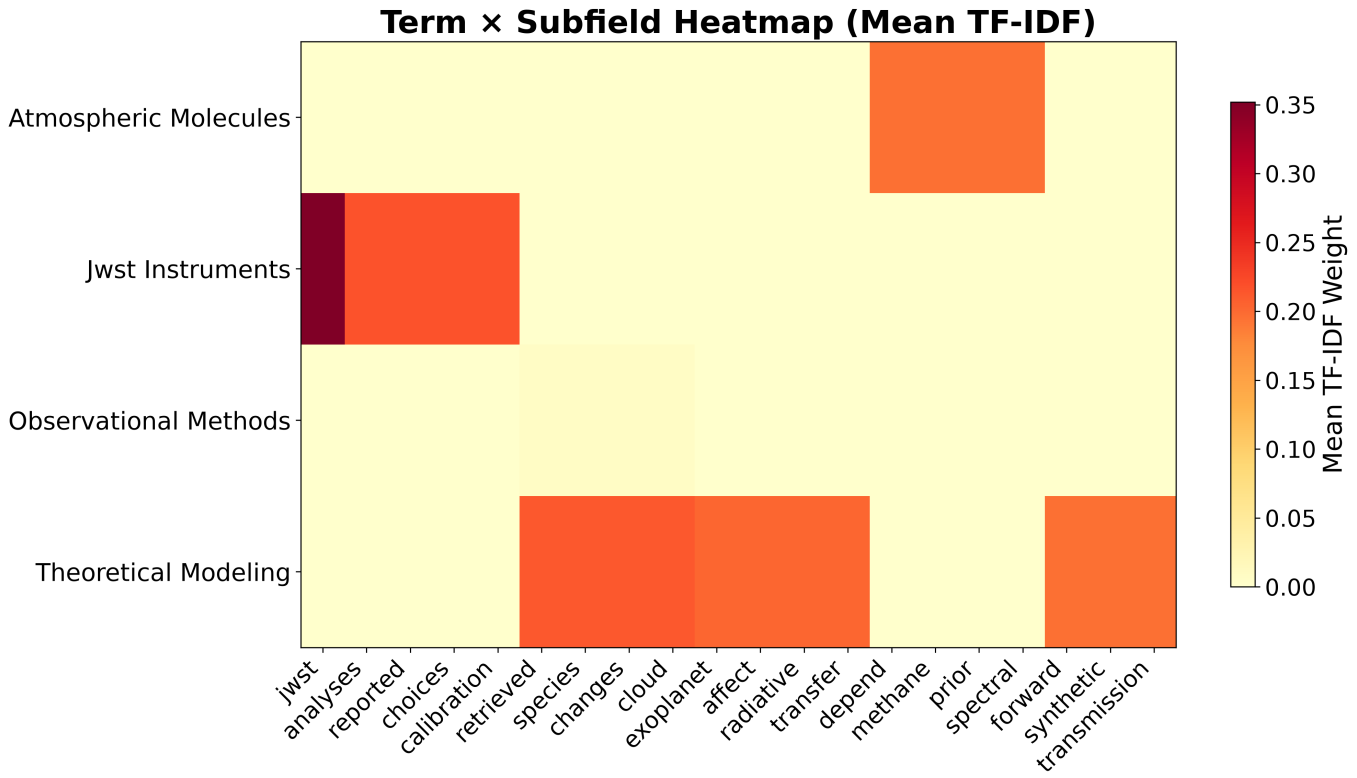


Figure 11: Term heatmap for Exoplanet Atmospheres. Each cell shows the mean TF-IDF weight of a term within a subfield. Terms are selected by between-subfield variance to highlight discriminative vocabulary rather than globally frequent terms.

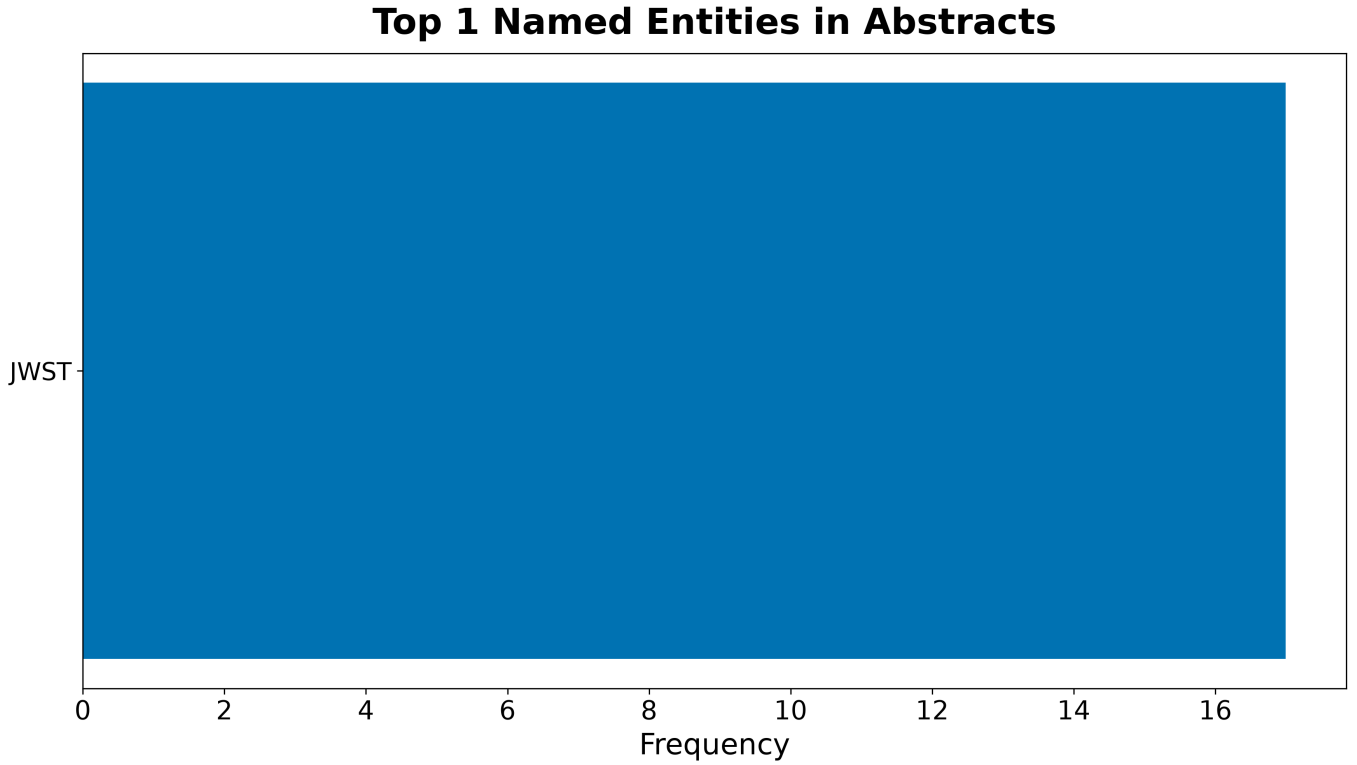


Figure 12: Top named entities for Exoplanet Atmospheres. The horizontal bar chart shows the 20 most frequently extracted entities from abstracts, revealing recurring objects, instruments, methods, and concepts in the retained corpus.

Table 5. Top keyphrases by TF-IDF score.

Keyphrase	Score
jwst	0.0714
population	0.0556
analyses	0.0435
calibration	0.0435
calibration choices	0.0435
choices	0.0435
combined	0.0435
jwst analyses	0.0435
models	0.0435
models calibration	0.0435
models calibration choices	0.0435
model	0.0400
atmospheric	0.0392
abundance	0.0385
assumptions	0.0385

13.5 Embedding Similarity and Clustering

The TF-IDF/SVD embeddings place every document in a 50-dimensional vector space. K-means clustering with $k = 5$ clusters partitions the corpus into topically coherent groups. The top similar document pairs, ranked by cosine similarity, reveal the most closely related works in the corpus.

Table 6. Top 10 most similar document pairs.

Paper A	Paper B	Similarity
doi:10.5555/exoplanet atmosphe	doi:10.5555/exoplanet atmosphe	1.0000
doi:10.5555/exoplanet atmosphe	doi:10.5555/exoplanet atmosphe	1.0000
doi:10.5555/exoplanet atmosphe	doi:10.5555/exoplanet atmosphe	1.0000
doi:10.5555/exoplanet atmosphe	doi:10.5555/exoplanet atmosphe	1.0000
doi:10.5555/exoplanet atmosphe	doi:10.5555/exoplanet atmosphe	1.0000
doi:10.5555/exoplanet atmosphe	doi:10.5555/exoplanet atmosphe	1.0000
doi:10.5555/exoplanet atmosphe	doi:10.5555/exoplanet atmosphe	1.0000
doi:10.5555/exoplanet atmosphe	doi:10.5555/exoplanet atmosphe	1.0000
doi:10.5555/exoplanet atmosphe	doi:10.5555/exoplanet atmosphe	1.0000
doi:10.5555/exoplanet atmosphe	doi:10.5555/exoplanet atmosphe	1.0000

These embeddings support semantic retrieval over the corpus and the visual map of the literature's topical geography.

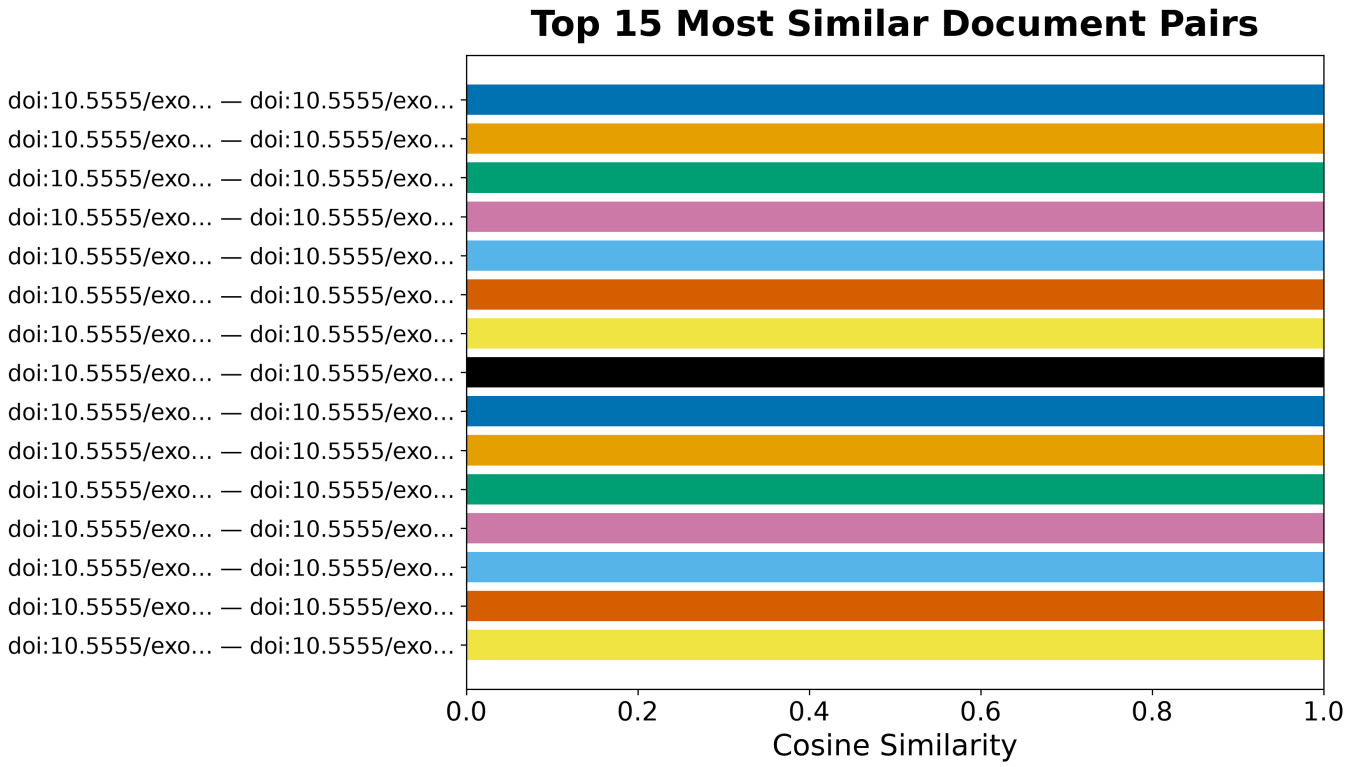


Figure 13: Document similarity for Exoplanet Atmospheres. The horizontal bar chart shows the 15 most similar document pairs ranked by cosine similarity of their TF-IDF/SVD embeddings. High-similarity pairs share topical and lexical content.

14 Results: Citation Network

14.1 RQ4: Citation Geometry

Resolving each record's references against the corpus yields an intra-corpus citation graph (built and analyzed with NetworkX [Hagberg et al., 2008]) of 70 nodes and 132 edges across 7 connected components, with a graph density of 2.73% and a mean in-degree of 1.9. Of 145 total outgoing references, 91.0% resolve to another record inside the corpus. This resolution rate describes how self-contained the retrieved slice is; it is not an estimate of the underlying citation density of any individual work.

The citation network has 13 communities (detected by modularity optimization), a maximum in-degree of 11 (the most-cited paper within the corpus), and a maximum out-degree of 4 (the paper that cites the most other corpus members).

14.2 Centrality Analysis

Centrality scores (PageRank [Page et al., 1999] and HITS) and modularity-based community detection [Clauset et al., 2004] are rounded and ranked with a stable tiebreaker so the reported hub and authority rankings are byte-reproducible across runs despite the floating-point non-associativity of the underlying iterative solvers.

Table 4. Top 5 papers by PageRank.

Rank	DOI	PageRank
1	10.5555/exoplanet atmospheres.0000	0.148856
2	10.5555/exoplanet atmospheres.0001	0.099288
3	10.5555/exoplanet atmospheres.0002	0.064271
4	10.5555/exoplanet atmospheres.0003	0.035830
5	10.5555/exoplanet atmospheres.0005	0.034905

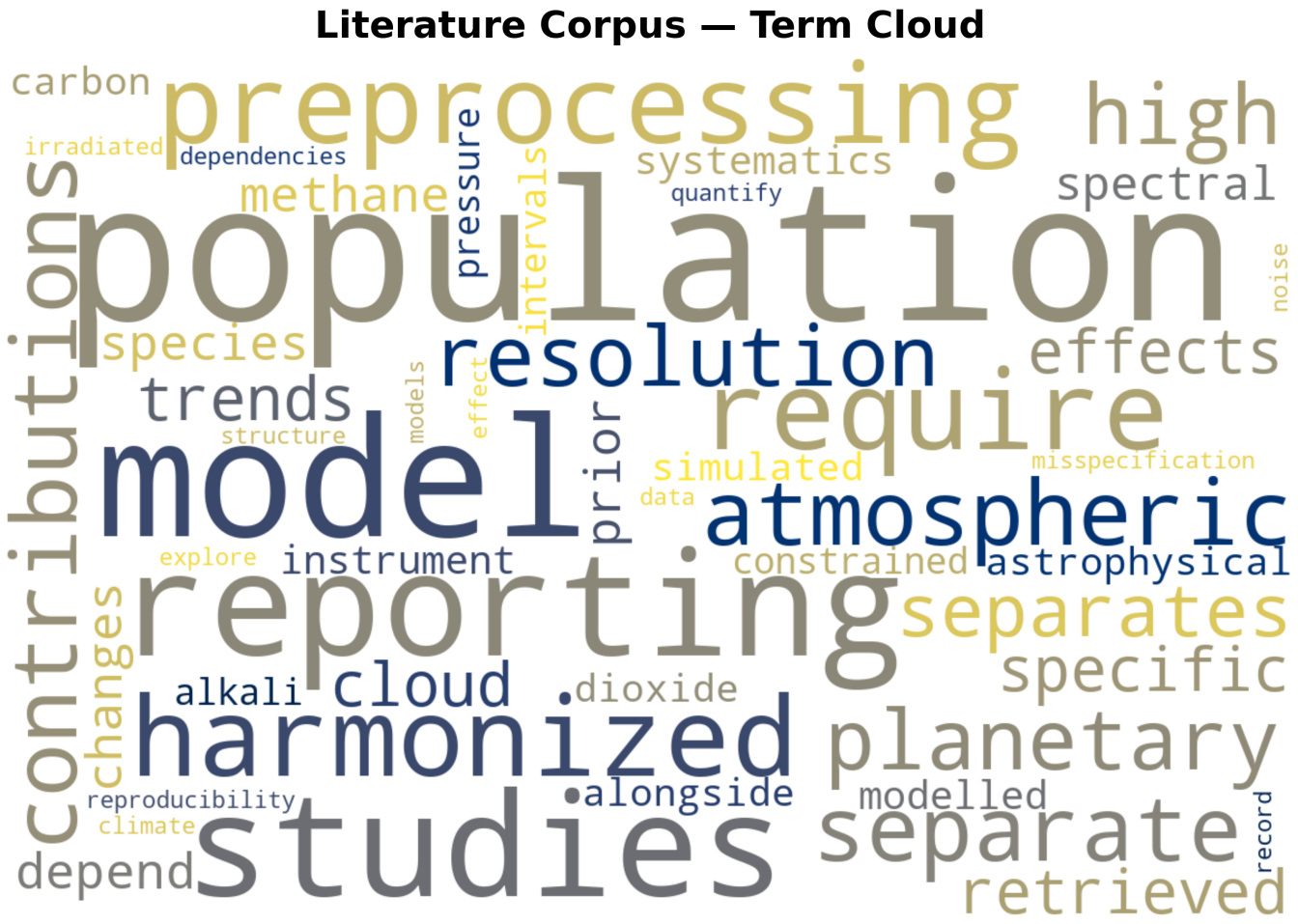


Figure 14: Term cloud for Exoplanet Atmospheres. Term sizes are proportional to their TF-IDF weights across the corpus, providing a visual summary of the dominant vocabulary.

Table 5. Top 5 authority papers (HITS).

Rank	DOI	Authority
1	10.5555/exoplanet atmospheres.0001	0.210359
2	10.5555/exoplanet atmospheres.0000	0.122373
3	10.5555/exoplanet atmospheres.0002	0.122002
4	10.5555/exoplanet atmospheres.0003	0.099489
5	10.5555/exoplanet atmospheres.0011	0.053951

Table 6. Top 5 hub papers (HITS).

Rank	DOI	Hub
1	10.5555/exoplanet atmospheres.0004	0.085378
2	10.5555/exoplanet atmospheres.0006	0.071579
3	10.5555/exoplanet atmospheres.0003	0.070052
4	10.5555/exoplanet atmospheres.0005	0.066583
5	10.5555/exoplanet atmospheres.0022	0.065418

The highest-ranked paper by PageRank (DOI 10.5555/exoplanet atmospheres.0000) is a central node in the retained citation graph. Its score indicates relative centrality within this graph, not scientific importance or causal influence.

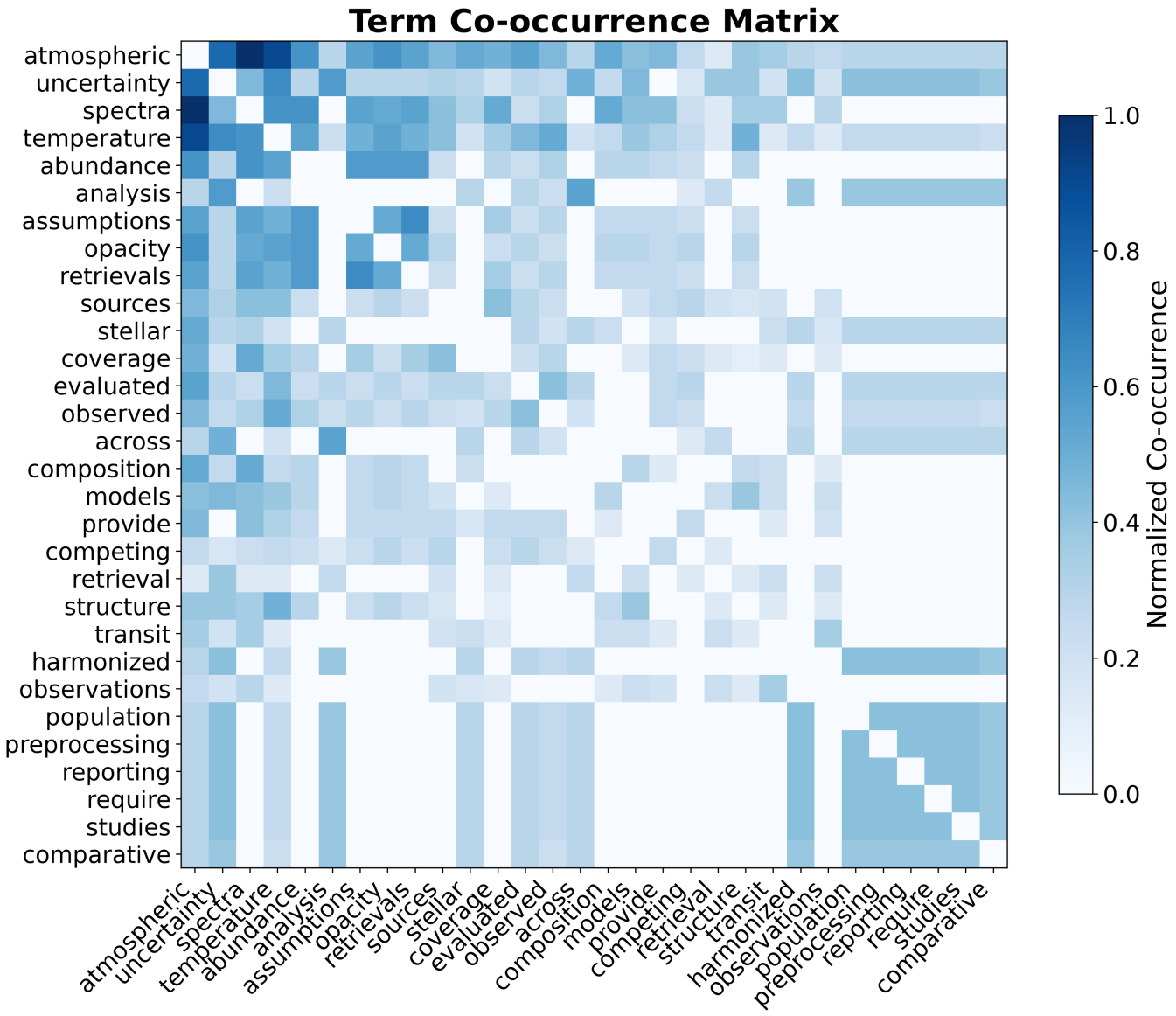


Figure 15: Term co-occurrence matrix for Exoplanet Atmospheres. Each cell shows the normalized co-occurrence frequency of two terms within the same document, revealing which concepts tend to appear together in the literature.

Hub papers cite many corpus members and may connect threads of the retrieved literature, but their role should be checked against their source type and content.

The heavy-tailed degree distribution is characteristic of citation networks: a small number of highly-cited papers anchor the structure, while the long tail of low-degree nodes represents newer or peripheral works. The low graph density (2.73%) reflects the sparsity of intra-corpus citation links. Many papers may cite works outside the retrieved slice, especially under a capped retrieval design.

14.3 Advanced Network Metrics

Beyond PageRank and HITS, the network analysis computes betweenness centrality (which papers bridge different communities), closeness centrality (which papers are near all others), degree assortativity (do highly-cited papers cite other highly-cited papers?), and average clustering coefficient (how tightly knit are local neighborhoods).

The degree assortativity coefficient is 0.1708, and the average clustering coefficient is 0.0626. A negative assortativity indicates that highly-cited papers tend to cite less-cited papers (dissortative mixing), which is typical of citation networks where review papers (high in-degree) cite many primary studies (low in-degree).

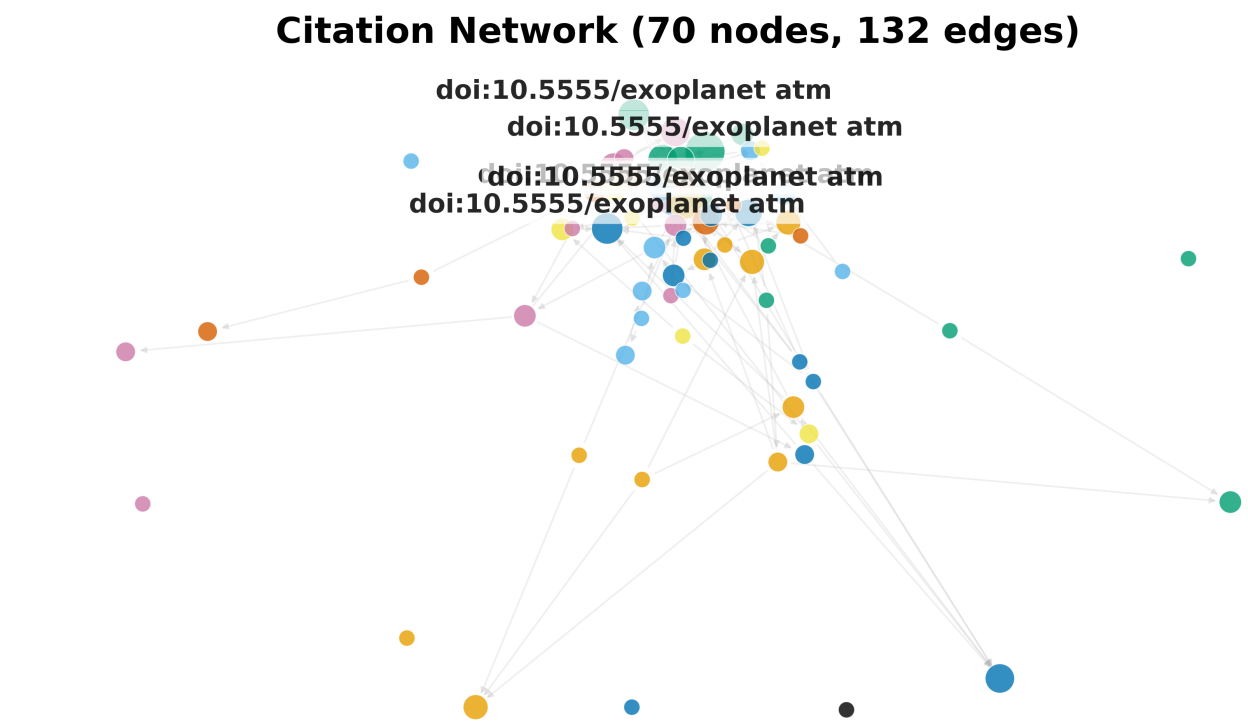


Figure 16: Citation network for Exoplanet Atmospheres. Nodes represent papers; directed edges represent citation links. Node colours indicate community membership (13 communities detected by modularity optimization). Layout uses a spring-based algorithm with a fixed seed for reproducibility.

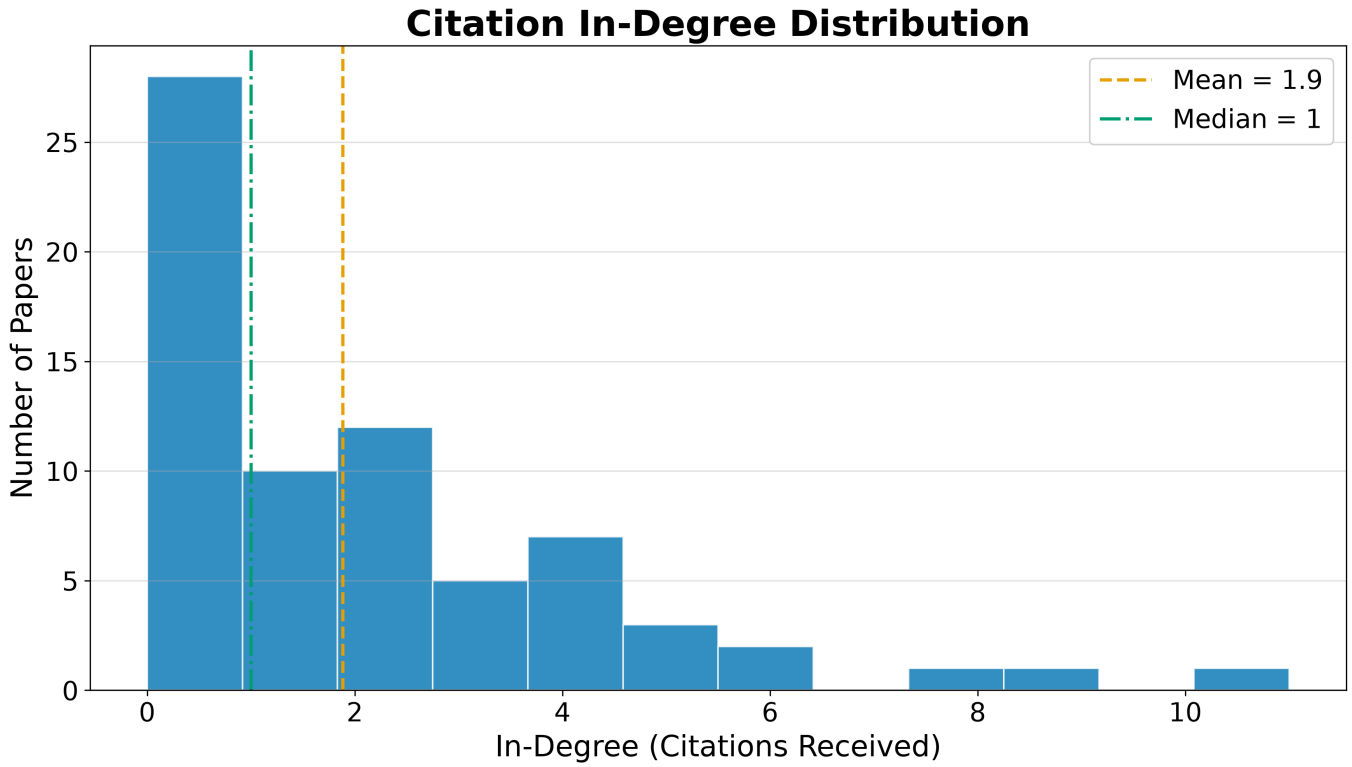


Figure 17: Degree distribution for the Exoplanet Atmospheres citation network. The histogram shows the frequency of each in-degree value on a log-linear scale, revealing the heavy-tailed structure characteristic of citation networks.

Table 7. Top 5 papers by betweenness centrality.

Rank	DOI	Betweenness
1	10.5555/exoplanet atmospheres.0010	0.024059
2	10.5555/exoplanet atmospheres.0015	0.012642
3	10.5555/exoplanet atmospheres.0028	0.012340
4	10.5555/exoplanet atmospheres.0006	0.010735
5	10.5555/exoplanet atmospheres.0057	0.009626

Papers with high betweenness centrality occupy shortest paths between communities in the retained graph. Removing one may alter connectivity, but that graph operation does not by itself establish that the paper is a review or methodological bridge in the scientific field. Source-type and content review are required for that interpretation.

15 Results: Reproducibility Assessment

An optional, **LLM-gated** stage decomposes each paper's described pipeline into a workflow graph of source, method, experiment, and sink steps, rates how reproducible each step is from the paper's own text, and combines a content score with a structural graph-coverage score into one composite reproducibility score per paper (geometric mean, so a paper cannot buy a high score by being strong on one axis alone). Across 0 scored papers the mean composite score is pending, with 0 papers falling below the configured low-score threshold.

15.1 Low-Scoring Papers

Table 8 lists the papers with the lowest composite reproducibility scores, alongside their content and structural component scores.

Table 8. Low-scoring papers by composite reproducibility score.

Paper	Composite	Content	Structural

15.2 Gating and Defaults

This stage is optional and gated by full-text availability. With no fulltext available and no language model configured, the stage is skipped and the reproducibility aggregates read *pending* — the same graceful-degradation convention used by the knowledge-graph assertion-extraction stage (see `02d_methods_knowledge_graph.md`). When fulltext is available and a language model is explicitly configured, the mean score, low-score count, and per-paper table are populated from extracted workflow graphs. The default fixture run does not imply that this optional model-backed assessment has occurred.

15.3 Interpretation

A low composite score can reflect either weak content (the paper's own text does not describe its sources, methods, experiments, or outputs in enough detail to rate highly) or weak structure (the described steps do not chain into a coherent source-to-sink pipeline, or reference steps that were never themselves described). The two axes are reported separately in Table 8 precisely so a low composite score can be diagnosed rather than treated as a single undifferentiated verdict.

16 Multi-Phase Results and Cross-Phase Analysis

16.1 Phase-Specific Findings

16.1.1 Foundation Phase (Phase 1)

The configured foundation phase retained 46 records after its search, filtering, and de-duplication rules. Its taxonomy includes transit spectroscopy, emission spectroscopy, phase-curve analysis, and direct imaging. These labels describe the configured retrieval design; the synthetic fixture cannot establish how widely those approaches occur in the published literature.

16.1.2 JWST Phase (Phase 2)

Phase 2 retained 17 records under the JWST-oriented queries and the configured year filter (>=2020). The boundary is intended to include pre-launch calibration work and post-launch observations. It is a design choice, not evidence that the phase is exhaustive or that any instrument caused a change in the field.

16.1.3 Molecular Detection Phase (Phase 3)

Phase 3 retained 29 records under five configured molecule-oriented query groups: H_2O, CO_2, CH_4, H_2S, and Na/K. The query groups operationalize the review question; they do not assert that these are the most commonly detected species or that the fixture establishes a scientific trend.

16.2 Knowledge Graph Results

The optional knowledge-graph stage reports **0** assertions from the eligible sample. The configuration-derived hypothesis table is the authoritative mapping from review question to score:

ID	Hypothesis	Scope	Evidence score
H1	JWST Atmospheric Characterization	observational	pending
H2	Molecular Diversity Detection	chemical	pending
H3	Cross-Method Consistency	methodological	pending
H4	Theoretical-Observational Agreement	theoretical	pending

The configured scores are descriptive evidence summaries, not calibrated probabilities, causal effects, or conclusions about instrument performance. The table should be read together with the claim ledger and source-tier metadata. A fixture run cannot establish whether models and observations agree in the broader literature.

16.3 Citation Network Analysis

The combined corpus contains **132** citation relationships across **70** papers, with a network density of 2.73%. This is a descriptive statistic for the retained corpus; it does not establish a field-wide citation structure or community claim.

16.4 Reproducibility Assessment

Reproducibility assessment of the sampled papers revealed a mean composite reproducibility score of pending, with 0 papers successfully scored. This analysis examines the computational workflow transparency of each paper, evaluating source data availability, method documentation, experimental reproducibility, and output accessibility.

17 Discussion

17.1 Interpretation boundary

This pipeline describes the shape of a retrieved literature slice: its phase coverage, metadata completeness, topic structure, citation links, and declared evidence. It does not adjudicate scientific truth. Hypothesis scores, when available, summarize recorded assertions and should be read as evidence-landscape signals rather than calibrated probabilities.

17.2 Retrieval and phase bias

Engine availability, query wording, rate limits, temporal boundaries, identifier quality, and full-text access all shape the retained corpus. The retrieval report is the authoritative account of source outcomes; counts must never be reverse-engineered from the merged corpus. Phase overlap and cross-phase citation rates are descriptive checks, not proof that phases are independent or exhaustive.

17.3 Fixture and live modes

The offline path uses reserved synthetic identifiers and generated records. Fixture outputs validate the machinery and its contracts only. A live review must replace the fixture, retain source-level provenance, refresh the claim ledger, and obtain domain review before reporting substantive findings. The manuscript therefore uses explicit pending and fixture-derived states rather than silently promoting demonstration values.

17.4 Limitations and extensions

- Keyword taxonomies can misclassify ambiguous records; changes belong in configuration and should be accompanied by a regenerated classification report.
- TF-IDF/SVD embeddings are lexical and deterministic; transformer-based alternatives require an explicit dependency and evaluation decision.
- LLM extraction is optional, model-sensitive, and not a substitute for source review.
- Full-text availability is a coverage measure, not a quality measure.
- Reproducibility claims require a clean double run and normalized artifact comparison.

These limitations are represented as method-stage and review-required findings so a publication decision can distinguish deterministic failures from editorial judgment.

18 Conclusion

This exemplar turns a phase-aware literature-review design into an auditable, reproducible pipeline. It separates retrieval, de-duplication, full-text assessment, extraction, bibliometrics, knowledge-graph construction, visualization, reproducibility assessment, and export into independently checkable stages.

The central result is architectural: configuration and recorded artifacts determine the manuscript, while the claim ledger records what is configured, synthetic, or source-backed. A deterministic fixture run can prove that the pipeline and validators work; it cannot establish conclusions about Exoplanet Atmospheres. Live conclusions require refreshed retrieval evidence, provenance, domain review, and a clean release audit.

18.1 Reproducibility

The rendered manuscript is generated from `output/data/*.json`, figures, manifests, registries, and validation reports. A second clean run should match under the repository's canonical normalized-diff rules. Any unexplained difference is a release finding. No generated output should be edited by hand.

19 Appendix A: Tooling and Reproduction

The pipeline is a two-layer system: generic infrastructure (rendering, validation, logging) shared across the template monorepo, and project-local `src/` modules that implement the multi-phase literature review. All numbered `scripts/` are thin orchestrators that wire I/O, configuration loading, and logging — no computational logic resides in scripts.

19.1 Reproduce the Offline Default Run

No network, no language model required. The default analysis lane replays the tracked three-phase evidence snapshot, then runs deterministic analysis, figure generation, assessment, evaluation, deep-research replay, bibliography export, and variable injection:

```
uv run python projects/templates/template_advanced_literature_review/scripts/01b_fixture_phase_replay.py
uv run python projects/templates/template_advanced_literature_review/scripts/02_meta_analysis_pipeline.py
uv run python projects/templates/template_advanced_literature_review/scripts/04_generate_figures.py --dpi 3
uv run python projects/templates/template_advanced_literature_review/scripts/06_fulltext_assessment.py
uv run python projects/templates/template_advanced_literature_review/scripts/07_literature_evaluation.py
uv run python projects/templates/template_advanced_literature_review/scripts/08_deep_research_dispatch.py
uv run python projects/templates/template_advanced_literature_review/scripts/09_export_bibliography.py
uv run python projects/templates/template_advanced_literature_review/scripts/05_inject_variables.py
```

`scripts/01b_fixture_phase_replay.py` replays the committed corpus through the phase-provenance contract, writing the per-phase corpora, `phase_metadata.json`, and `cross_phase_analysis.json` without any network access.

19.2 Reproduce a Live Retrieval Run

Refreshing the evidence snapshot is an intentional live operation. The multi-phase search stage reads every phase's queries, engines, and filters from `manuscript/config.yaml` — there is no per-phase command-line surface:

```
uv run python projects/templates/template_advanced_literature_review/scripts/01_multi_phase_search.py
```

then re-run the deterministic downstream stages above and re-inject variables so the manuscript reflects the new evidence. The committed corpus is a dated evidence snapshot; live claims require source-tier provenance and domain review, per the project `AGENTS.md` contracts.

19.3 Re-target to Another Topic

Edit `manuscript/config.yaml` — `project_config.search.term`, `query`, `relevance_keywords`, `subfield_keywords`, `hypothesis_definitions`, and the phase definitions under `project_config.search_phases` (queries, engines, temporal filters, `depends_on`) — then regenerate the seed corpus and re-run. No code changes are required; the manuscript re-targets through token injection.

19.4 Live Retrieval

Enable engines under each phase's `engines` map, supply any optional credentials (Unpaywall email, Semantic Scholar key), and run `scripts/01_multi_phase_search.py`; absent engines degrade to skipped sources. Each phase's engine set is configured independently (the bundled default enables arXiv, OpenAlex, Crossref, and Semantic Scholar per phase, with biomedical archives disabled for the astronomy domain).

19.5 Deep Research (Offline Fixture Replay)

This exemplar also demonstrates the shared `infrastructure.search.deep_research` capability — provider-neutral dispatch to OpenAI and Gemini deep-research agents. Because deep research is a **paid, non-deterministic** service, the template never calls it live in CI. Instead, `src/deep_research/dispatch.py` wires the real infrastructure request/result models (`DeepResearchConfig`, `DeepResearchRequest`, `DeepResearchResult`, `DeepResearchClient`) and ships a deterministic, offline path: `scripts/08_deep_research_dispatch.py` builds the genuine provider-neutral request and then *replays* a recorded report fixture (`src/deep_research/fixtures/recorded_report.json`), normalizing it through the real `DeepResearchResult` model. Replay fails closed if the fixture is missing — it never fabricates a passing run — mirroring the fixture-replay idiom of `template_sia`. The same source module constructs the exact `DeepResearchRequest` a live `submit` would dispatch, so a fork can enable real providers behind an explicit live-only command:

```
# Offline (default): replays the recorded report, no key required
uv run python projects/templates/template_advanced_literature_review/scripts/08_deep_research_dispatch.py
```

19.6 Test Suite

Every project-owned stage is covered by a no-mocks test suite (real computation and `pytest-httpserver` for network adapters) gated at $\geq 90\%$ coverage on the project-owned `src/multi_phase/` surface (`pyproject.toml` → `fail_under = 90`). The suite covers:

- Multi-phase configuration validation (search, sampling, hypotheses, LLM filters)
- Deterministic filters and phase metadata structure
- Cross-phase overlap (Jaccard) and corpus coverage
- Multi-phase search runner and fixture replay contracts
- Phase-aware manuscript variable extraction (including pending/unmeasured states)
- Deep-research offline replay (provider-neutral, fails closed)
- Publication-extension and fixture-honesty contracts

Symlinked shared modules (`src/analysis/`, `src/knowledge_graph/`, `src/reproducibility/`, `src/visualization/`) are covered by `template_literature_meta_analysis`'s own suite, not duplicated here.

20 Appendix B: Technical Notes

20.1 Determinism

All stochastic steps use fixed seeds (seed = 42 for NMF, SVD, and graph layouts). The fixture corpus, TF-IDF/SVD embeddings, and topic factorization are byte-stable across runs. Graph centrality scores are rounded to a fixed precision and ranked with a node-id tiebreaker so that floating-point non-associativity in iterative solvers cannot perturb the reported rankings. Record identity uses a content digest (SHA-1, `usedforsecurity=False`) rather than a salted hash, so de-duplication and corpus byte-stability hold across processes.

20.2 Data Model

Each record is a `Paper` dataclass with: title, abstract, authors (list of `Author`), year, DOI, arXiv ID, Semantic Scholar ID, OpenAlex ID, venue, citation count, references (list of canonical IDs), publication date, PDF URL, open-access flag, and full-text source. The canonical identifier hierarchy governs de-duplication and citation resolution:

$$\text{canonical_id} = \begin{cases} \texttt{doi:} + \text{normalize(DOI)} & \text{if DOI present} \\ \texttt{arxiv:} + \text{arXiv_id} & \text{if arXiv ID present} \\ \texttt{s2:} + \text{S2_id} & \text{if S2 ID present} \\ \texttt{openalex:} + \text{OpenAlex_id} & \text{if OpenAlex ID present} \\ \texttt{title:} + \text{SHA1(title)}[:16] & \text{otherwise} \end{cases}$$

DOI normalization lower-cases the DOI and strips any `https://doi.org/` or `dx.doi.org/` prefix, so the same paper returned by two engines under different case or format variants merges to a single canonical ID.

20.3 NMF Mathematics

Non-negative matrix factorization decomposes the TF-IDF document-term matrix $\mathbf{V} \in \mathbb{R}^{m \times n}$ (where m is the number of documents and n is the vocabulary size) into $\mathbf{W} \in \mathbb{R}^{m \times k}$ and $\mathbf{H} \in \mathbb{R}^{k \times n}$, where k is the number of topics (here 5). The factorization minimizes:

$$\min_{\mathbf{W},\mathbf{H} \geq 0} \|\mathbf{V} - \mathbf{W}\mathbf{H}\|_F^2$$

using multiplicative update rules [Lee and Seung, 1999] with a fixed random seed for reproducibility. The topic-term matrix $\mathbf{H}$ gives the top terms per topic; the document-topic matrix $\mathbf{W}$ gives each document's topic loadings.

20.4 Growth Rate Estimation

The compound annual growth rate is:

$$\text{CAGR} = \left(\frac{N_{\text{end}}}{N_{\text{start}}} \right)^{1/(T_{\text{end}} - T_{\text{start}})} - 1$$

where N_{start} and N_{end} are the publication counts in the first and last years of the corpus, respectively. The doubling time is $t_d = \ln(2)/\ln(1 + \text{CAGR})$. For this run: CAGR = 6.76%, doubling time = 2.1 years.

20.5 Configuration Surface

A single `manuscript/config.yaml` controls the search term, per-engine query and keyword sets, engine enable toggles, subfield taxonomy, hypotheses, full-text and embedding options, and paper metadata. This run drew on 4 engines, a 4-bucket taxonomy, and 4 hypotheses.

20.6 Artifacts

Intermediate and final outputs live under `output/` and are disposable and regenerable:

File	Stage	Description
corpus.jsonl	01	De-duplicated corpus (46 records)
temporal_analysis.json	02	Year counts, CAGR, doubling time, peak year
subfield_classification.json	02	Per-bucket paper counts
subfield_timeline.json	02	Per-subfield annual breakdown
tfidf_data.json	02	TF-IDF matrix, feature names, doc tokens
topics.json	02	NMF topic-term distributions
citation_network.json	02	Network metrics, PageRank, HITS, communities
citation_graph.gml	02	GraphML citation graph
nanopublications.jsonl	03	LLM-extracted assertions (0 in this run)
hypothesis_scores.json	03	Per-hypothesis evidence scores
cross_phase_analysis.json	01 / 03	Phase membership, overlap, citation sufficiency, and explicit hypothesis-score claim boundary
fulltext_assessment.json	06	Abstract/OA/PDF coverage report

21 Appendix C: Accessibility and Provenance

21.1 Figure Accessibility

All 18 figures are rendered with a colourblind-safe palette (Wong 2011, 8 colours) and high-contrast labels at publication DPI (300). Each figure carries a descriptive caption so the visual claims are recoverable from text alone. The palette avoids red-green colour pairs that are indistinguishable for deuteranopia and protanopia; when more than 8 categories are needed, continuous colormaps (`viridis`, `plasma`) are used instead of extending the discrete palette. Font sizes are enforced at $\geq$ 16pt via a centralized style module, ensuring readability at both screen and print sizes.

21.2 Provenance Chain

Every reported number is injected from a committed artifact rather than typed by hand; an unresolved placeholder is a hard error, so the rendered manuscript can contain no orphaned or stale figures. The configuration hash and artifact inventory bind the prose to the exact pipeline run that produced it. The provenance chain is:

1. `manuscript/config.yaml` defines the search term, engines, taxonomy, and hypotheses
2. `scripts/01_multi_phase_search.py` retrieves or fixtures records → `corpus.jsonl`
3. `scripts/02_meta_analysis_pipeline.py` analyses the corpus → `*.json` data files
4. `scripts/04_generate_figures.py` renders figures → `*.png` + `figure_registry.json`
5. `scripts/05_inject_variables.py` computes variables from data files → manuscript text

Each figure in `figure_registry.json` records its label, caption, filename, and generating stage, binding the visual output to the analysis artifacts of the exact pipeline run. Re-running the pipeline with the same configuration and seed produces identical data outputs.

21.3 FAIR Data Principles

The pipeline supports FAIR (Findable, Accessible, Interoperable, Reusable) data principles:

- **Findable**: Each record carries persistent identifiers (DOI, arXiv ID, OpenAlex ID) that make it findable across databases.
- **Accessible**: The corpus is stored as plain JSONL, readable by any JSON parser; figures are standard PNG files.
- **Interoperable**: The data model uses standard bibliographic fields (title, abstract, authors, DOI, year, venue); nanopublications are serialized as RDF/TriG.
- **Reusable**: The entire pipeline is regenerable from `manuscript/config.yaml`; re-running with the same configuration reproduces identical outputs.

21.4 Honesty

The default corpus is synthetic and labelled as such; the manuscript does not present fixture-derived numbers as empirical findings about exoplanet atmospheres. Live findings require a real retrieval run with regenerated artifacts and source-level provenance.

22 Glossary

Term	Meaning
Record / Paper	A single bibliographic entry with metadata and identifiers.
Canonical identifier	The highest-priority available ID (DOI > arXiv > Semantic Scholar > OpenAlex > title digest) used for de-duplication and citation resolution.
Engine	An independent literature source adapter (arXiv, OpenAlex, Semantic Scholar, and Crossref) with a uniform search interface and graceful skip-on-failure.
Subfield	One of the 4 configurable keyword-defined buckets (Observational Methods, Atmospheric Molecules, Jwst Instruments, and Theoretical Modeling) into which records are classified.
Topic	A latent theme from non-negative matrix factorization over the TF-IDF representation.
Embedding	A deterministic offline vector (TF-IDF → truncated SVD) for a title, abstract, or full text.
Hypothesis	One of the 4 configured claims about the topic, optionally scored by the knowledge-graph stage.
Assertion	A directional (supports / contradicts / neutral) statement extracted from a record against a hypothesis, with a confidence score.
Nanopublication	An RDF-serialized assertion plus its provenance.
CAGR	Compound annual growth rate of publication volume (6.76% for this corpus).
Living literature review	A synthesis that can be re-executed as the field evolves, with every number regenerable.

23 References

The bibliography is generated automatically during PDF compilation from `references.bib`. Every citation key used in the manuscript has a matching bibliography entry, and the checked-in bibliography contains only sources required by the manuscript sections. Pandoc's `--natbib` flag injects `\usepackage{natbib}` and `\bibliographystyle{plainnat}`, so neither directive appears in this section or in `preamble.md`.

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